Reminder

We would like to draw your attention to the provisions of European and national competition law which prohibit the discussion of competitively sensitive topics, such as prices or discounts and to otherwise exchange sensitive company data during association meetings. Forecasts of future business development are particularly critical in this regard and can only be made in aggregated form through VDMA.

Similarly, it is prohibited to agree on common behaviours in an industry and/or to pass resolutions or make arrangements in this respect. Such actions may be subject to severe fines that have to be paid by the association and its member companies. This is why these rules must be adhered to without exception.

For any queries, please do not hesitate to approach the responsible officer in your company, your contact person at VDMA or VDMA Legal Services. Additional information can be found in the “Do’s and Don’ts” (Code of Conduct). We are happy to send you a copy of the VDMA Compliance Programme.

Getting Started

Getting Started

- **Overview of OPC Foundation, OPC UA and IJT**
  - https://opcfoundation.org/about/opc-technologies/opc-ua/
  - https://opcfoundation.org/markets-collaboration/IJT/

- **OPC UA IJT Group Presentation:** Refer to the **OPC UA IJT Group Presentation.pdf/pptx.**
  - https://github.com/umati/UA-for-Industrial-Joining-Technologies/tree/main/IJT_Documents

- **Specifications/Online Reference**
  - **Joining:** https://reference.opcfoundation.org/nodesets?u=http://opcfoundation.org/UA/IJT/Base/
  - **Tightening:** https://reference.opcfoundation.org/nodesets?u=http://opcfoundation.org/UA/IJT/Tightening/

- **OPC UA IJT Prototypes/Reference Implementations**
  - https://github.com/umati/UA-for-Industrial-Joining-Technologies

  - **OPC UA Generic Client: LINK**
    - This the **primary generic** test client which is free of charge and can be used to understand any OPC UA concepts.
  - **IJT Server Simulator: LINK**
    - The functionality in the simulator is like the OPC UA Server on a Device. Refer to the document in the above link for **usage** instructions.
  - **IJT Web Client: LINK**
    - It is a reference IJT Web Client for understanding the concepts visually.
  - **IJT Console Client: LINK**
    - It is a simple python-based client to get results. It can also be used as a base to integrate in the automated test framework.
  - **IJT C# Client: LINK**
    - It is a simple C# Client based on OPC Foundation C#.NET SDK for example usage.
  - **IJT Test Client: LINK**
    - It is a base test client for testing OPC UA IJT Server features. It is a base testing framework, and continuous work is in progress.

OPC UA Overview (IEC 62541)

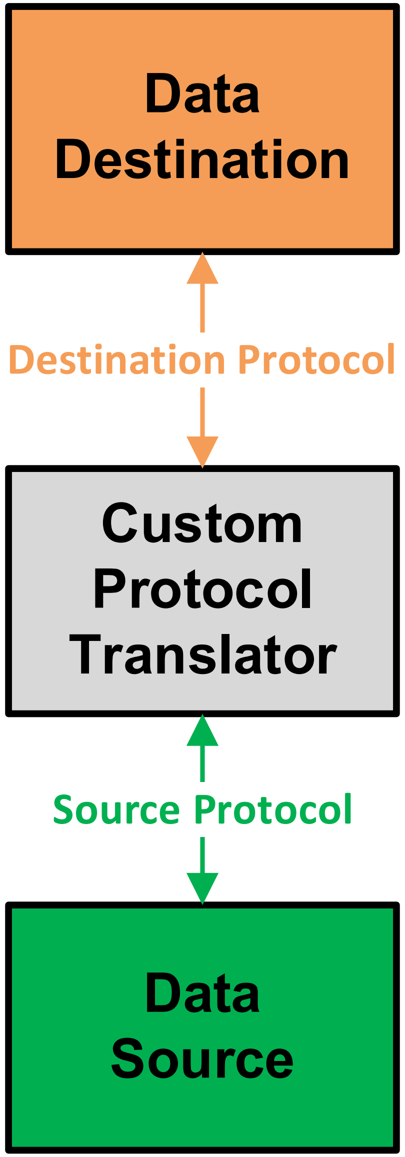
Problem Statement

- **I4.0 Challenges for Communication**
  - Interoperability
  - Security
  - …

Traditional Solution – Custom Protocol Converter – Before OPC

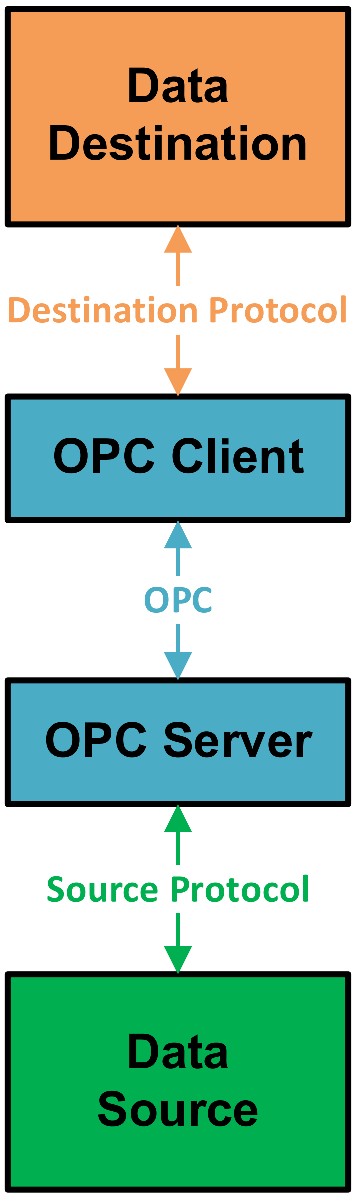
- Custom software is built through an extensive collaboration of both system's vendors.
  - Enabled tightly controlled communication between the source and destination regardless of complexity.

- **But it has the following limitations:**
  - Vendor Specific
  - Costly to produce
  - Costly to manage
  - Proprietary Data Paths
  - Little Integration
  - Each application had custom drivers



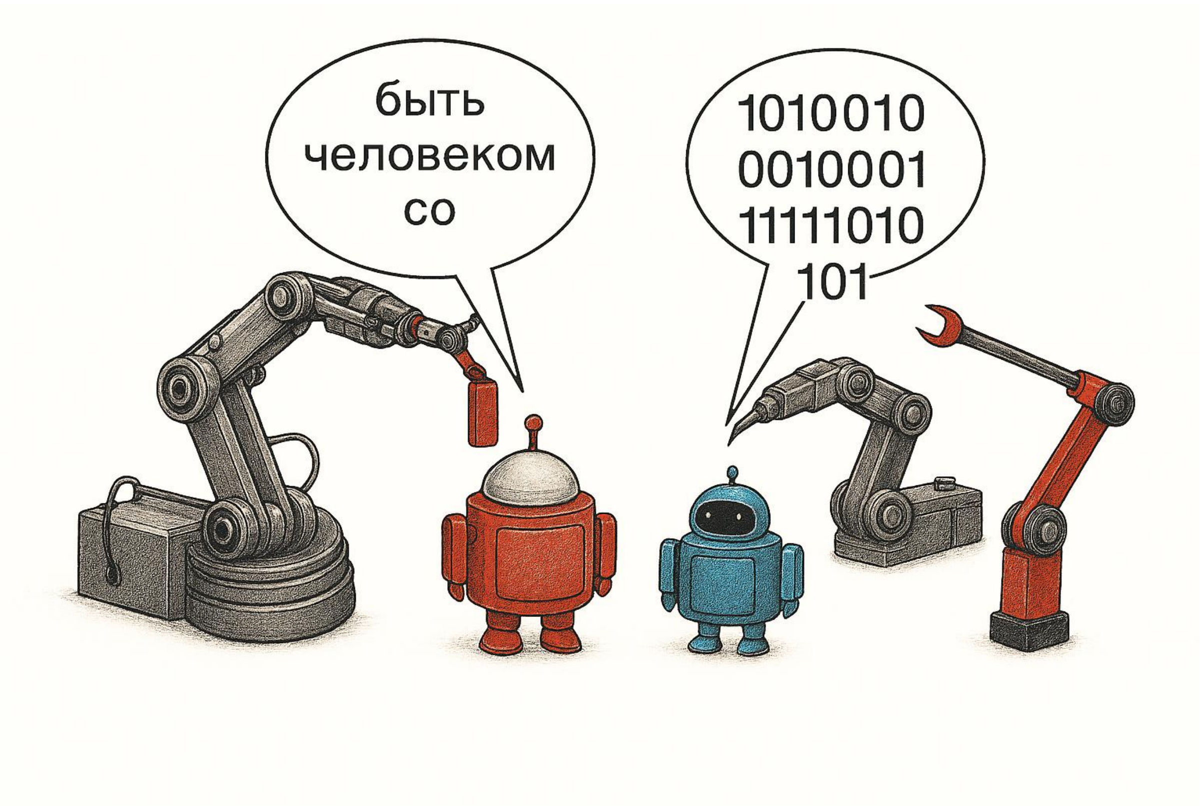
OPC Connectivity – OPC Classic Data Path

- All OPC Data Paths will consist of these four components.
- More complex architectures will consist of many OPC Data Paths.
- Understanding the OPC Data Path will enable understanding all OPC Solutions.
  - Architecting
  - Implementing
  - Troubleshooting

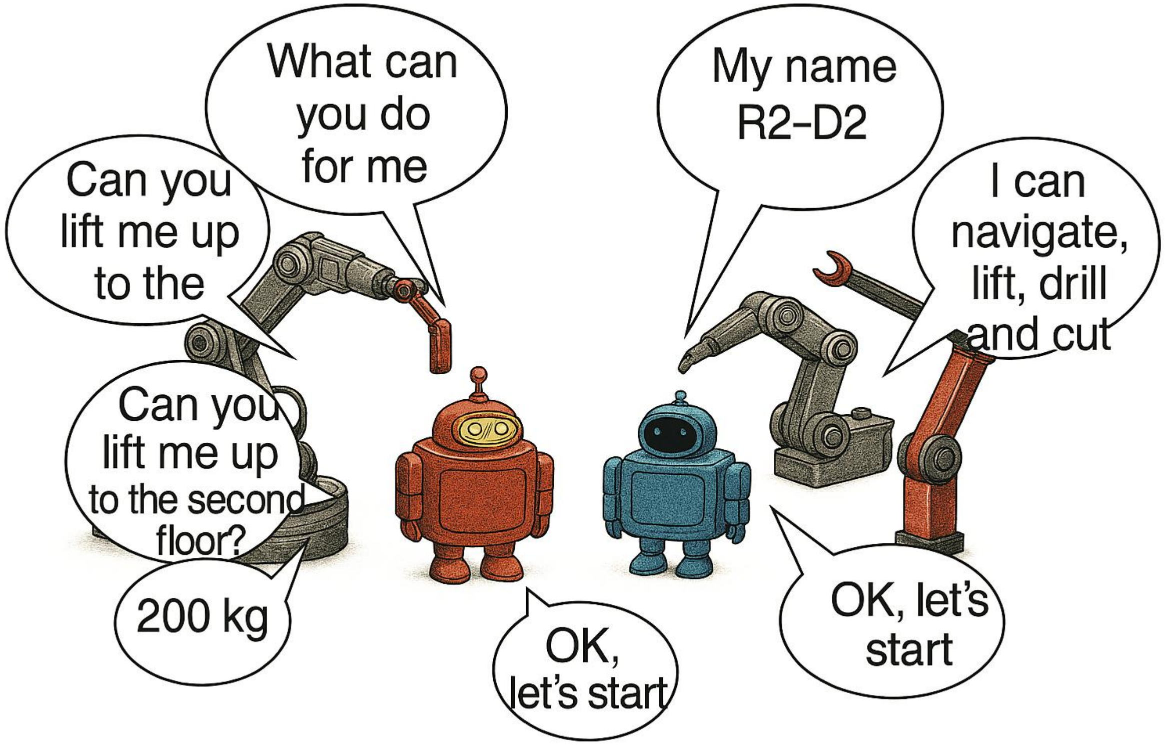


Interoperability Use Case

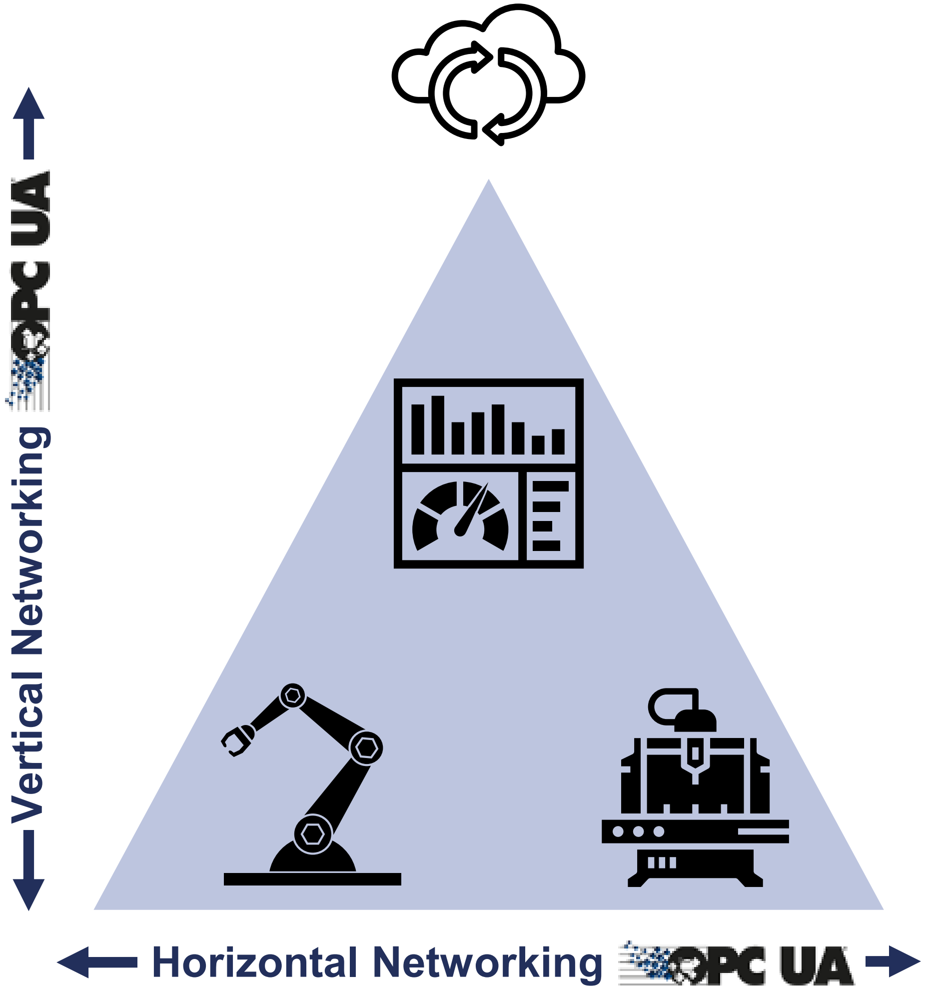
Without Interoperability



With Interoperability

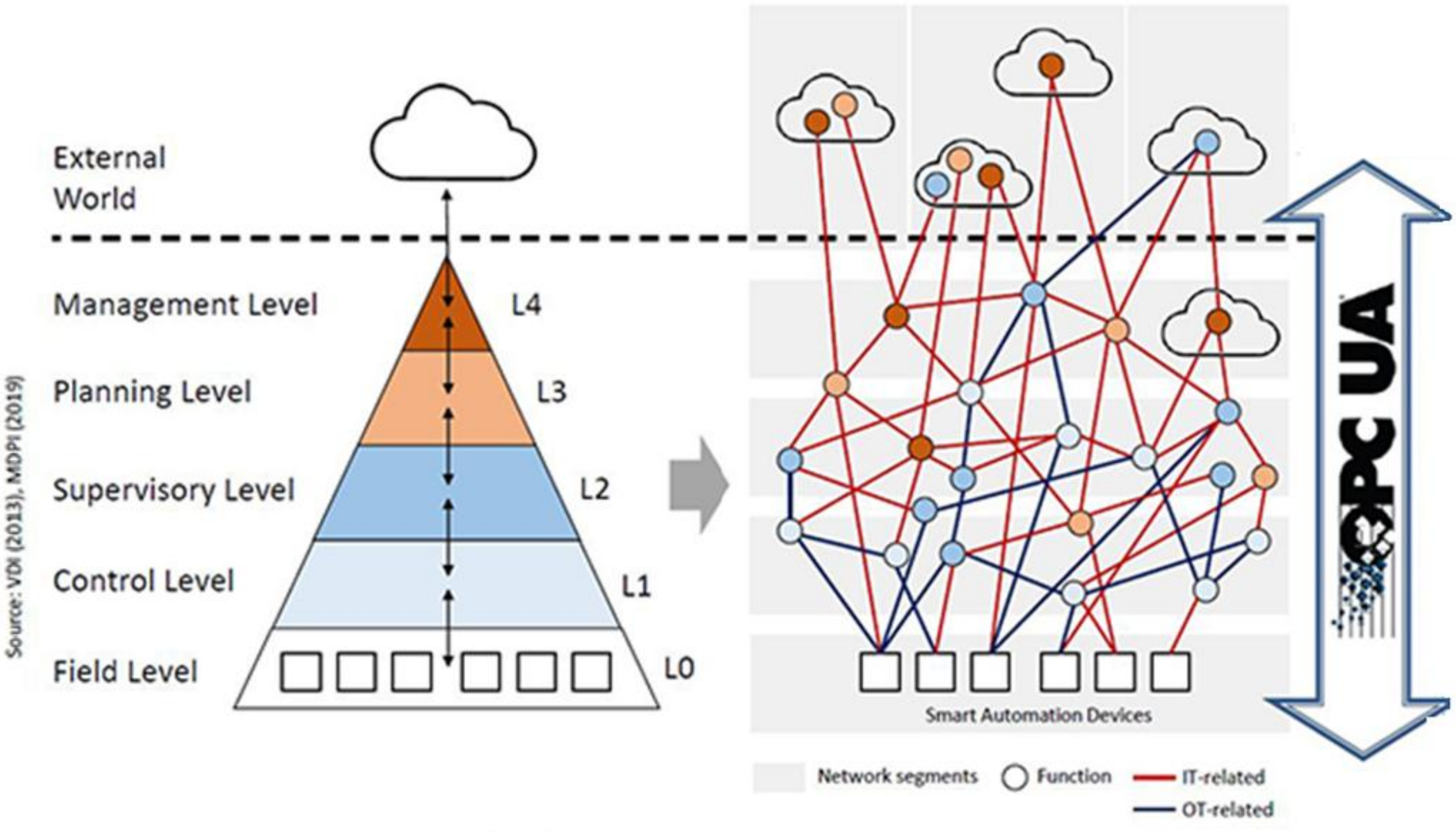


OPC UA – Field to Cloud



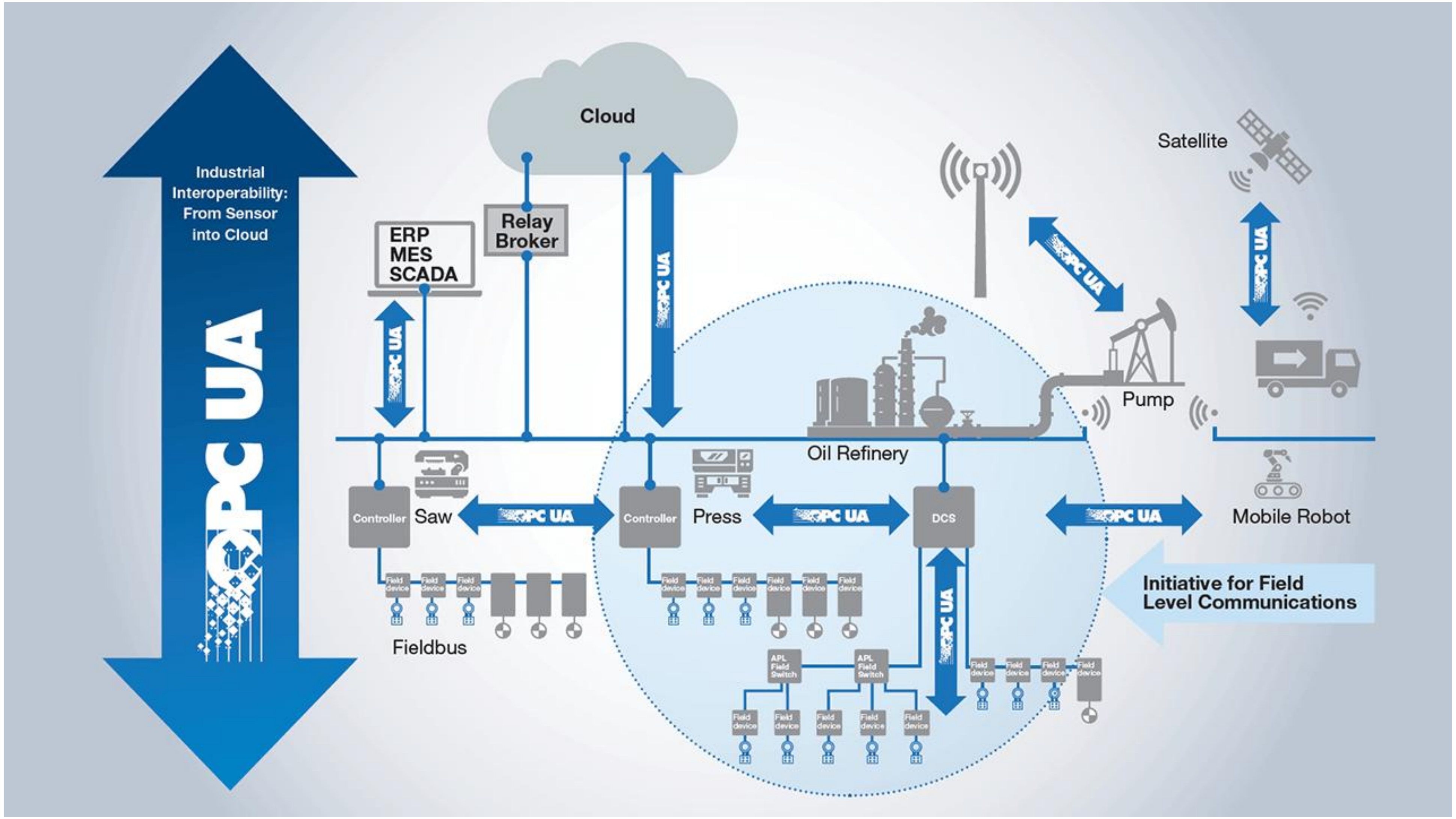
- Open source
- Security
- Two transport mechanisms with various protocols
- Scalable
- Global acceptance
- Semantic information models

From Automation Pyramid to Information Network



Source: OPC Foundation

OPC UA Factory Network Example



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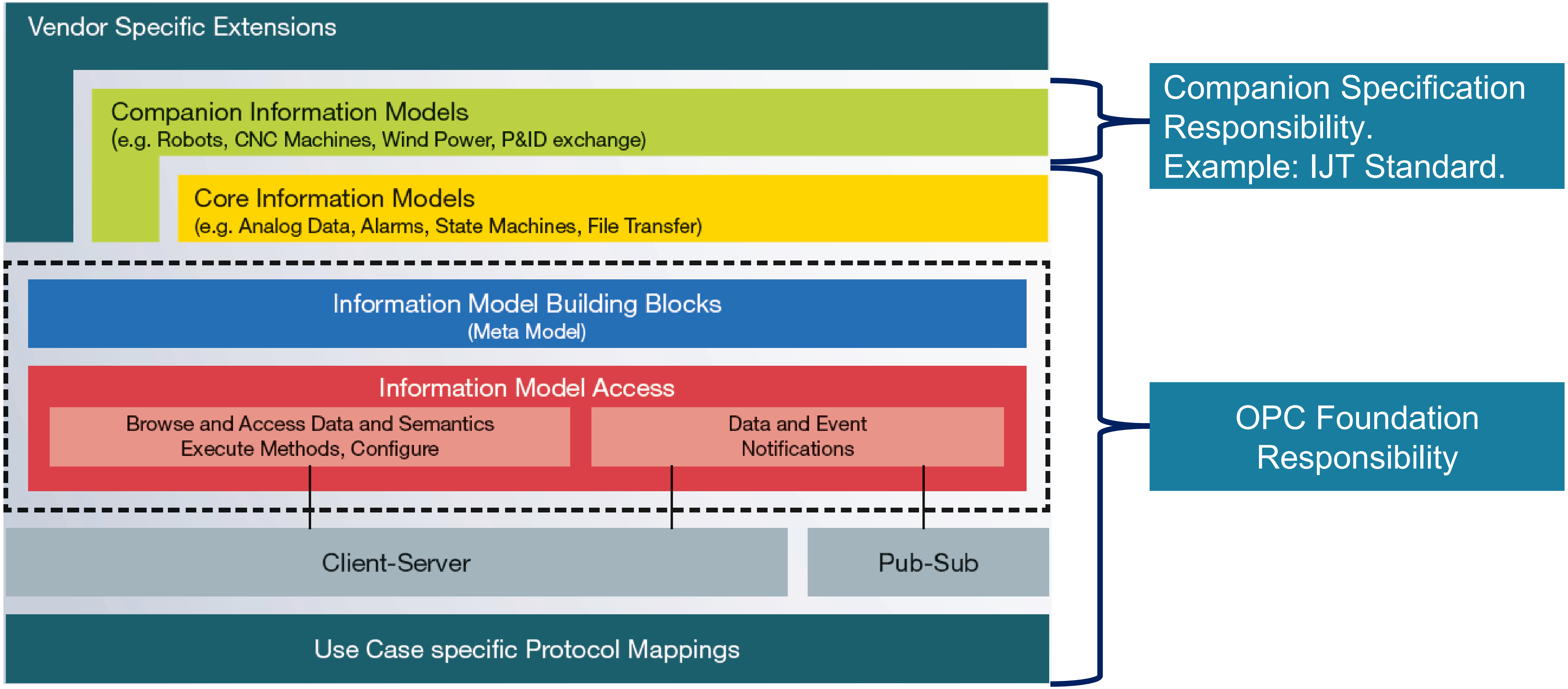
OPC UA Layered Architecture



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OPC UA Deployment Options

Platform Independence

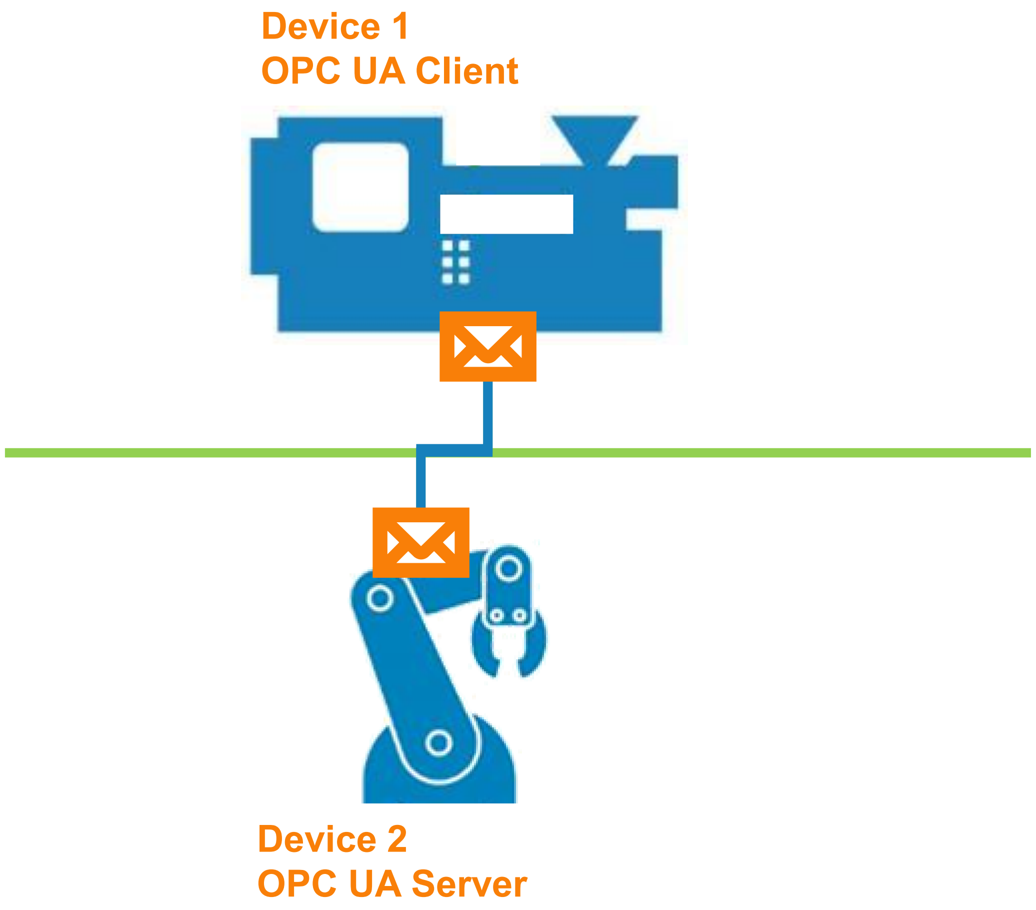
Hardware

Operating Systems

Real Time
Operating
System

UNIX

OPC UA Client/Server Architecture



CLIENT/SERVER

- Client accesses information from the server via a permanently configured connection

EXAMPLE: REQUEST → RESPONSE

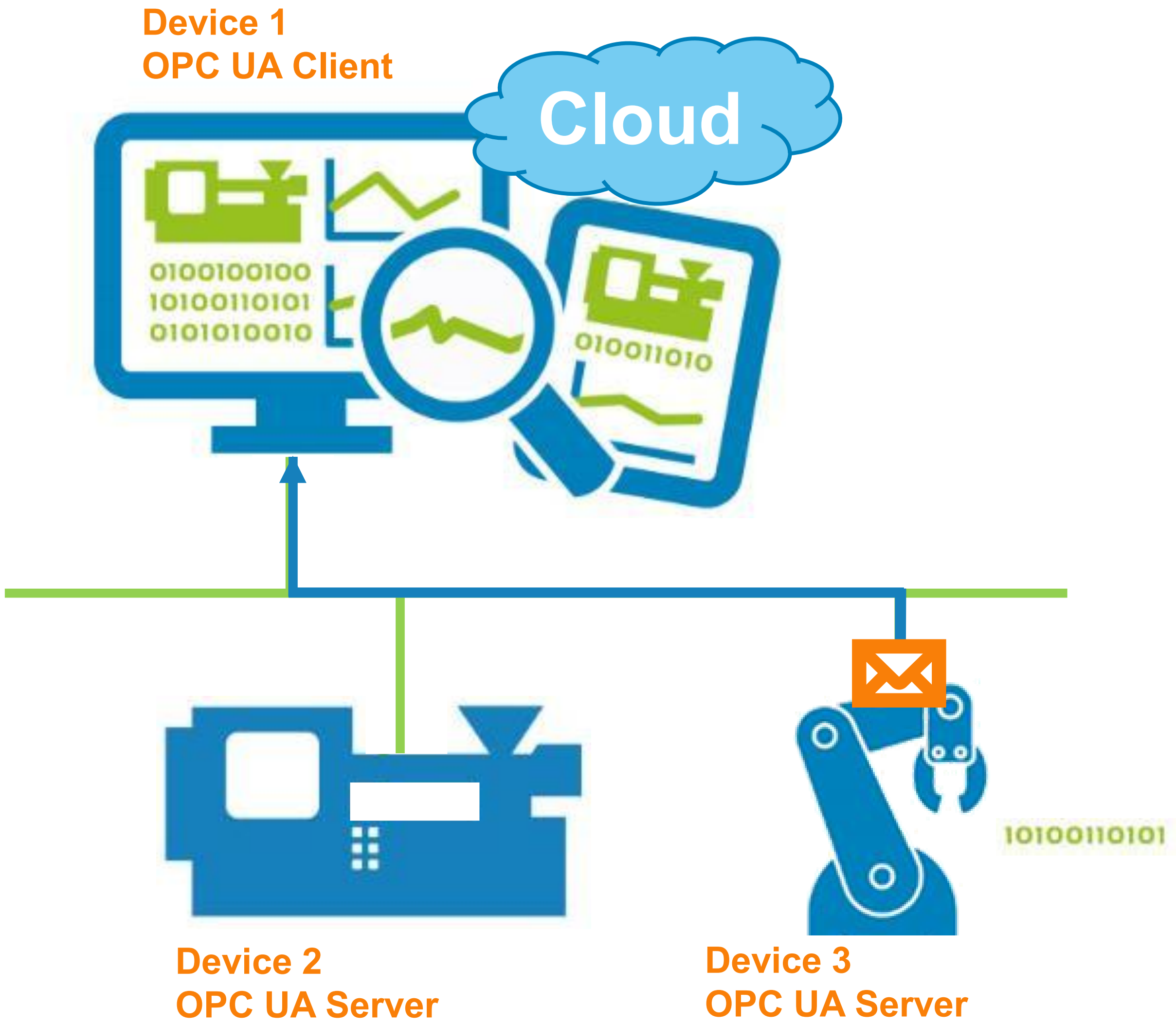
- Device 1 requests information from Device 2
- Device 2 replies to Device 1

ANALOGY

- Letter service with registered mail and advice of receipt.

OPC UA – Client/Server Architecture

Communication via Subscription Notify



CLIENT/SERVER

- Client accesses information from the server via a permanently configured connection

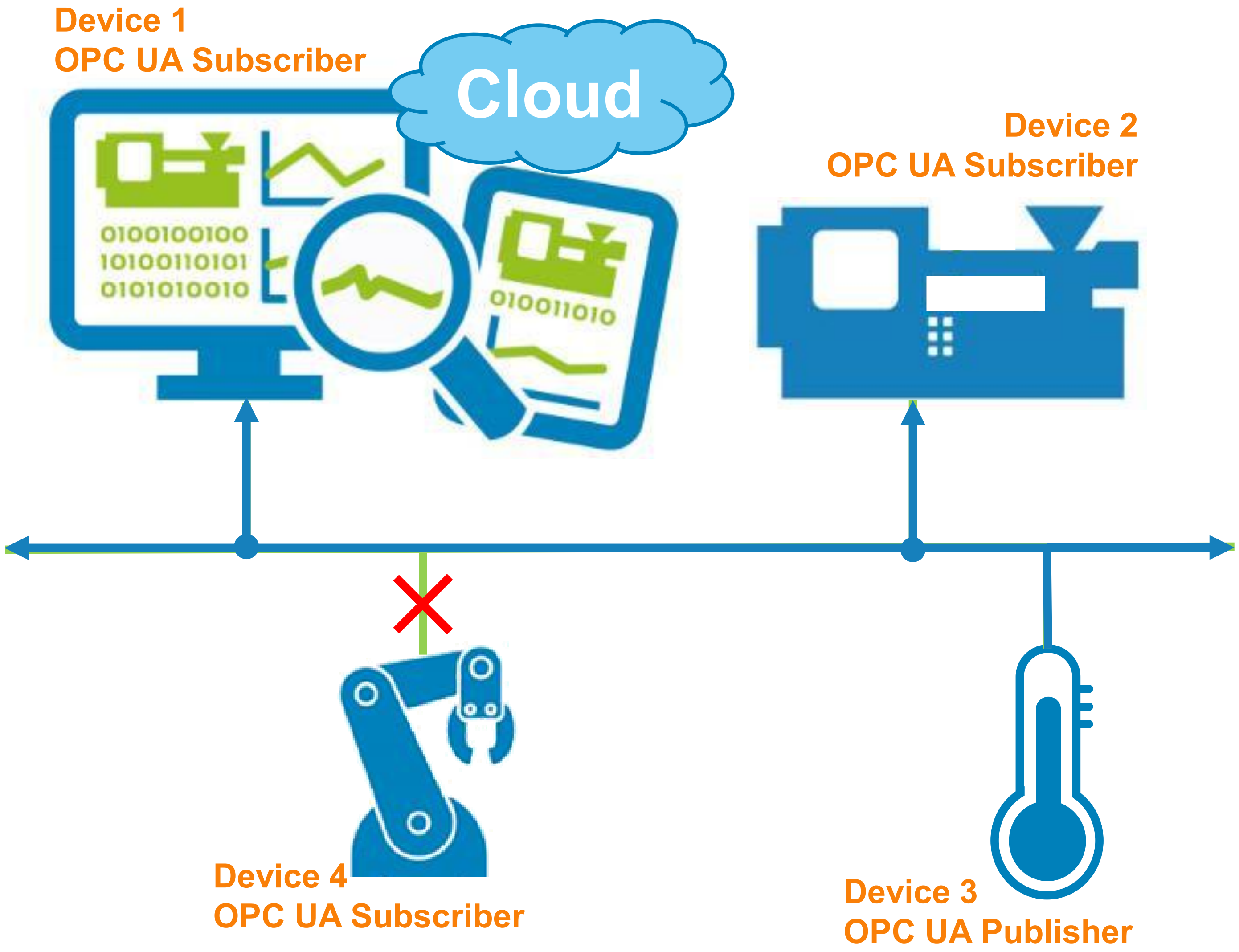
EXAMPLE: SUBSCRIPTION → NOTIFY

- Device 1 subscribes to the information from Device 3
- Device 3 informs Device 1 as soon as information changes

ANALOGY

- Registration to football live ticker with feedback on special events via SMS

OPC UA Pub/Sub Architecture



PUBLISHER/SUBSCRIBER

- Publisher sends to unknown subscribers without a fixed connection
- Application example:
  - 1:n - Sensor publishes data used by different systems

EXAMPLE: PUB/SUB

- Device 3 continuously sends information to the network
- Device 1 and 2 have subscribed to the information
- Device 4 has not subscribed to the information

ANALOGY

Setting a radio frequency

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Object-oriented, flexible, and extendable.

Domain-specific models.

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PLUS – Discovery and Security

OPC UA Security

OPC UA Security

Security

Security is incorporated into OPC UA at all levels!

- Confidentiality and integrity of communication is only part of the solution

- End-to-End Communication Security
- User Authentication
- Roles and role Management
- Audit logging
- Certificate management infrastructure

Source: OPC Foundation

OPC UA Security Objectives – Trusted Information (CIA Triad)

CONFIDENTIALITY

- Keep private data private.

INTEGRITY

- Data remains unchanged from sender to receiver.

AVAILABILITY

- Availability, by restricting the message size and returning no security related codes

OPC UA Security Objectives – Access Control (AAA Framework)

Authentication

- Proof of Identity based on something the entity is, has or knows.

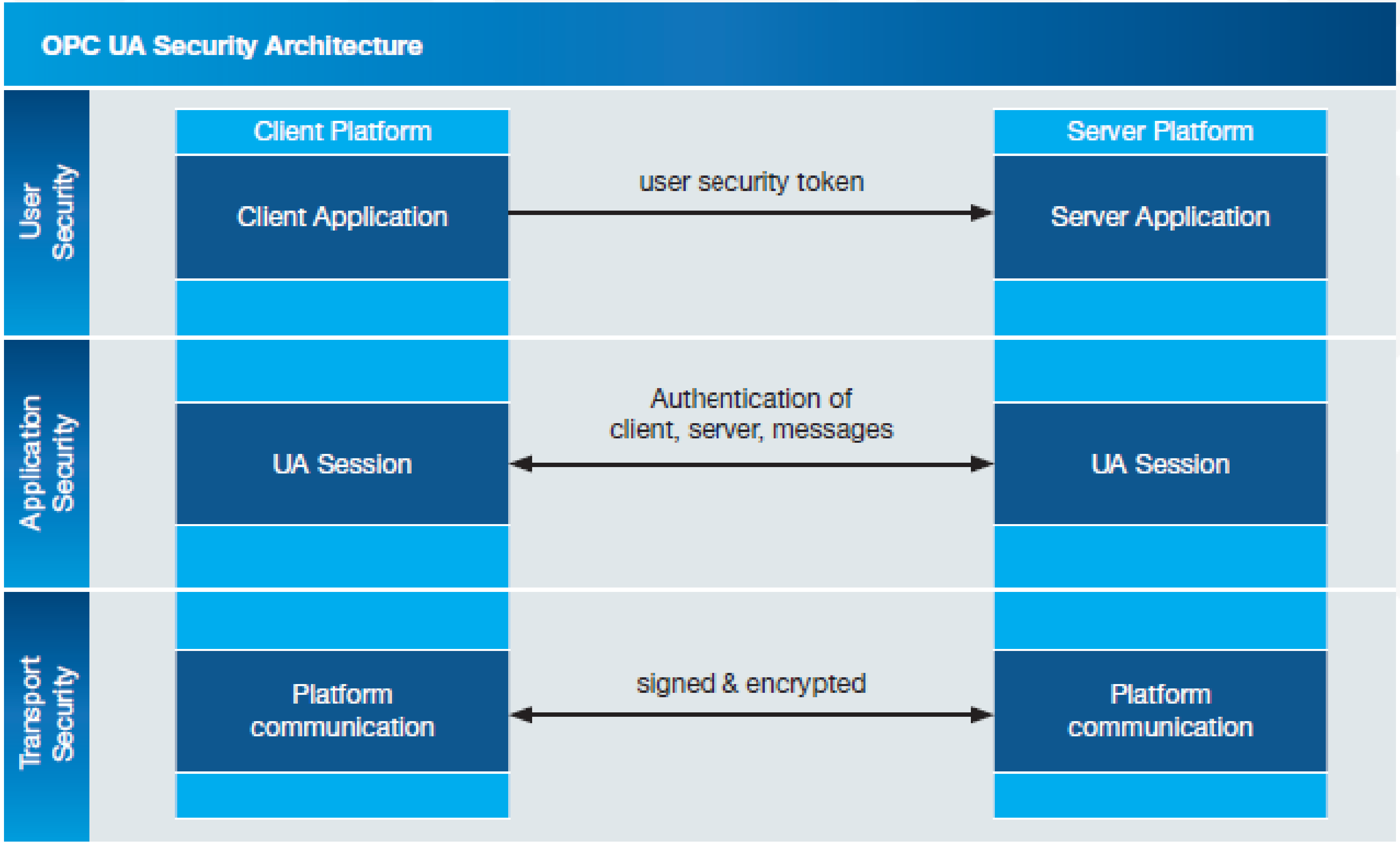
Authorization

- Not specifically in the OPC UA specification but is vendors choice to implement.

Auditability

- Provide evidence that the system functions as specified and identify initiators of certain actions.

Security Architecture



Source: OPC Foundation

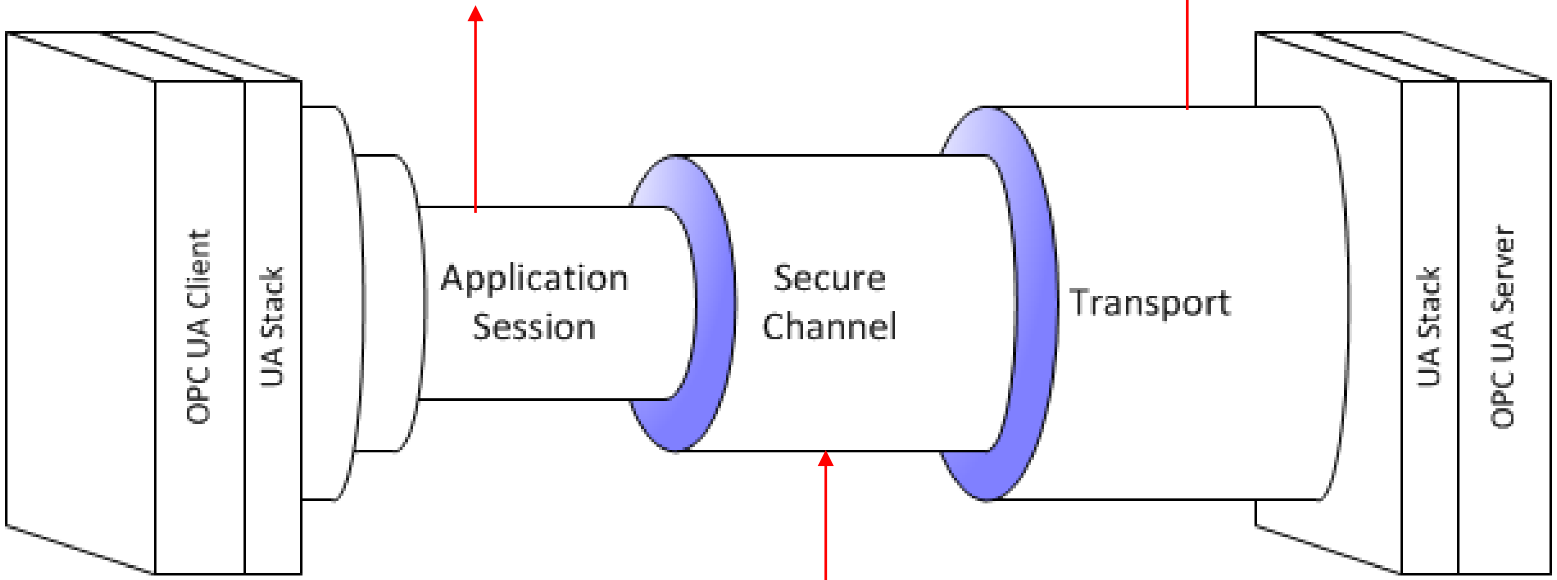
The Connection Process

Secure Session

- Application layer connection.
- The session uses the algorithms defined for the channel and assigns an identity to the connection.

Transport Layer

- Protocol determined by the UA Stack.
- Security not required, but may be implemented by the protocol, i.e. HTTPS.



Secure Channel

- Long-running logical connection between single client and server.
- Ensures *Confidentiality* and *Integrity* of the messages exchanged.
- Supports three *Security Modes*: None, Sign, and SignAndEncrypt.
- Latest security algorithms.

Source: OPC Foundation

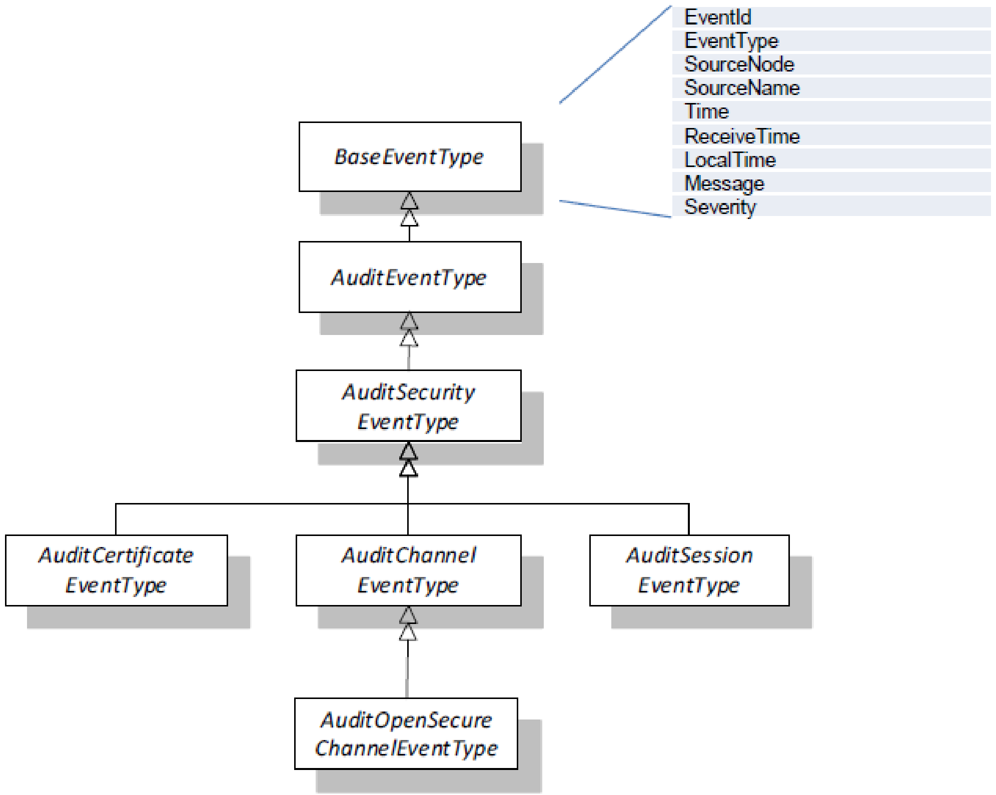
Security

Security Goals	Authenti-cation	Integrity	Accounting	Confidenti-ality	Availability	Authoriza-tion
Threats	Spoofing	Tampering	Non-Repudiation	Information Disclosure	Denial of Service	Escalation of Privileges

Auditing and Events

Auditing and Events

- AuditEvents are Events that are generated as a result of an action taken on the Server by a Client of the Server.
- AuditSecurityEvents are Events related to Security such as validating a Certificate or UserIdentityToken.
- AuditEvents may be reported vis Subscriptions, PubSub or via non-UA mechanisms such as SYSLOG.
- The structure and semantics of the events is the same no matter how they are reported.



Source: OPC Foundation

Roles and Permissions

Roles And Permissions

- Roles are associated with a Session
- Permissions are associated with a combination of a Role and a Node.
- A Session will have access to a Permission if the one or more of its Roles has the Permission.
- Mapping Rules are used to determine which Roles are available for a Session

Role	Mapping Rules	Description
Anonymous	Identities = Anonymous Applications = Endpoints =	An identity mapping rule that specifies the Role applies to anonymous users.
AuthenticatedUser	Identities = AuthenticatedUser Applications = Endpoints =	An identity mapping rule that specifies the Role applies to authenticated users.
Operator1	Identities = User with name 'Joe' Applications = urn:OperatorStation1 Endpoints =	An identity mapping rule that specifies specific users that have access to the Role with a application rule that restricts access to a single Client application.
Operator2	Identities = Users with name 'Joe' or 'Ann' Applications = urn:OperatorStation2 Endpoints =	An identity mapping rule that specifies specific users that have access to the Role with a application rule that restricts access to a single Client application.
Supervisor	Identities = User with name 'Root' Applications = Endpoints =	An identity mapping rule that specifies specific users that have access to the Role
Administrator	Identities = User with name 'Root' Applications = Endpoints = opc.tcp://127.0.0.1:48000	An identity mapping rule that specifies specific users that have access to the Role when they connect via a specific Endpoint.

Source: OPC Foundation

OPC UA Global Discovery Server (GDS)

Problem: How do we ensure certificate management in a factory?

- OPC UA specifies a Global Discovery Server

OPC UA Global Discovery Server (GDS)

- Provides the necessary infrastructure to provide enterprise-wide administration of OPC UA Servers.
- Handles discovery of registered OPC UA servers.
- **…but also handles certificate management.**

Major organizations have implementations of OPC UA GDS.

- Example: OPC Vault from Microsoft

OPC UA Companion Specifications

Purpose of Companion Specification

- OPC UA Companion Specifications are developed for various reasons:
  - To publish specific information models (e.g., for specific industries, specific devices, specific use cases).
  - To specify how to use OPC UA in specific environments.
  - **Examples:**
    - OPC UA standard for Robotics (OPC 40010), Vision Systems (OPC 40100), Joining Systems (OPC 40450 and 40451), etc.

- **New Information Models** can be created based on the OPC UA Data Model and eventually derived from OPC UA **Base** Information Models.
  - Companion specifications of such Information Models are often called “**Industry standard models**” because they typically address a dedicated industry problem.
  - The synergy of the OPC UA infrastructure to exchange such industry information models enables interoperability at the semantic level.

Harmonization and Reuse

- The purpose of harmonization is to ensure that several companion standards reuse common building blocks. Several use cases could be common for various domains.
  - **Examples:** Asset Identification, Asset States, Machinery Result, etc.

- OPC Foundation and partner organizations such as VDMA enable the reuse of common elements across domains.

- There are several OPC Harmonization groups where representatives of different companion specifications participate and harmonize the information models.
  - **Examples:** Device Integration, Machinery Building Blocks, Machinery Result Transfer, Asset Management Basics, Relative Spatial Location, and many more...

- Any new companion specification planned to be developed is recommended to be reviewed in the OPC Harmonization Group.

- This approach helps in avoiding reinvention of the wheel.

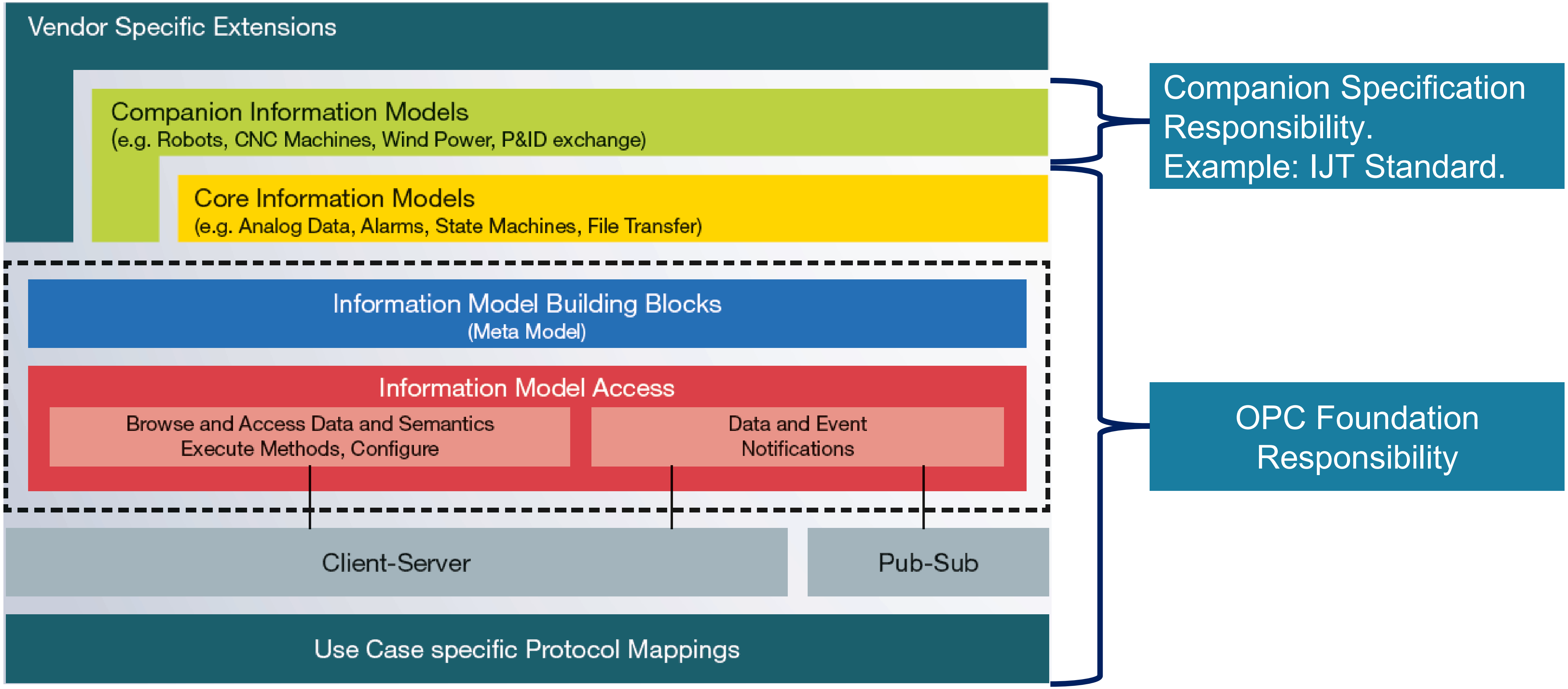
Ways to produce companion specifications

OPC Foundation differentiates **three** ways of producing companion specifications:

- **INTERNAL:** These are specifications created by OPC-internal working groups.
- **JOINT:** These are specifications that are created in a joint working group between the OPC Foundation and another organization.
  - These joint specifications represent the majority. The JOINT working group program is defined here: https://opcfoundation.org/joint-working-groups/
  - **Example:** IJT Working Groping is a Joint Working Group of OPC Foundation, VDMA and various vendors. VDMA hosts the IJT Working Group.
- **EXTERNAL:** Companion specifications can also be created independent of the OPC Foundation.

OPC UA Stack/SDK

OPC UA Layered Architecture



Source: OPC Foundation

OPC UA Stack and SDKs

- OPC UA SDKs are available in various programming languages.
  - Ansi C, C++, C#, Java, Python, NodeOPCUA, etc.

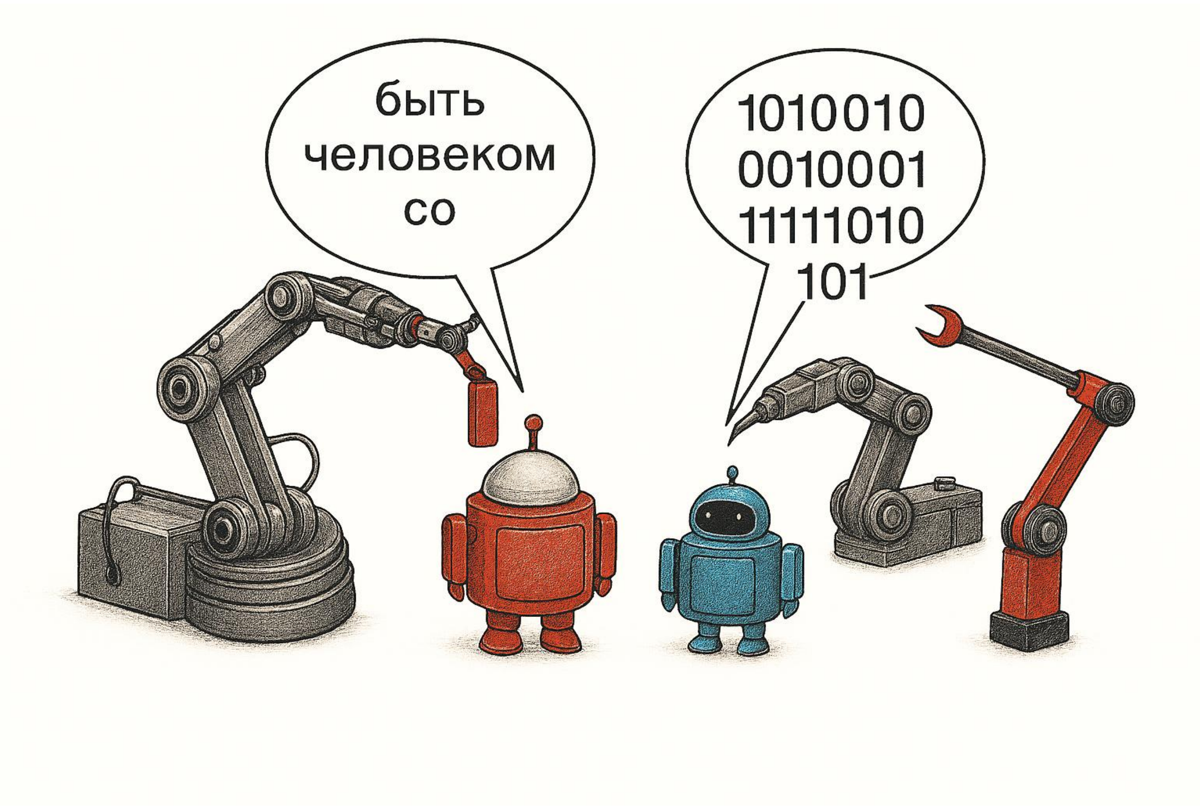
- The core functionality of OPC UA is handled as part of OPC UA SDK.

- OPC Foundation and other organizations provide various frameworks.
  - Open Source
  - Commercial

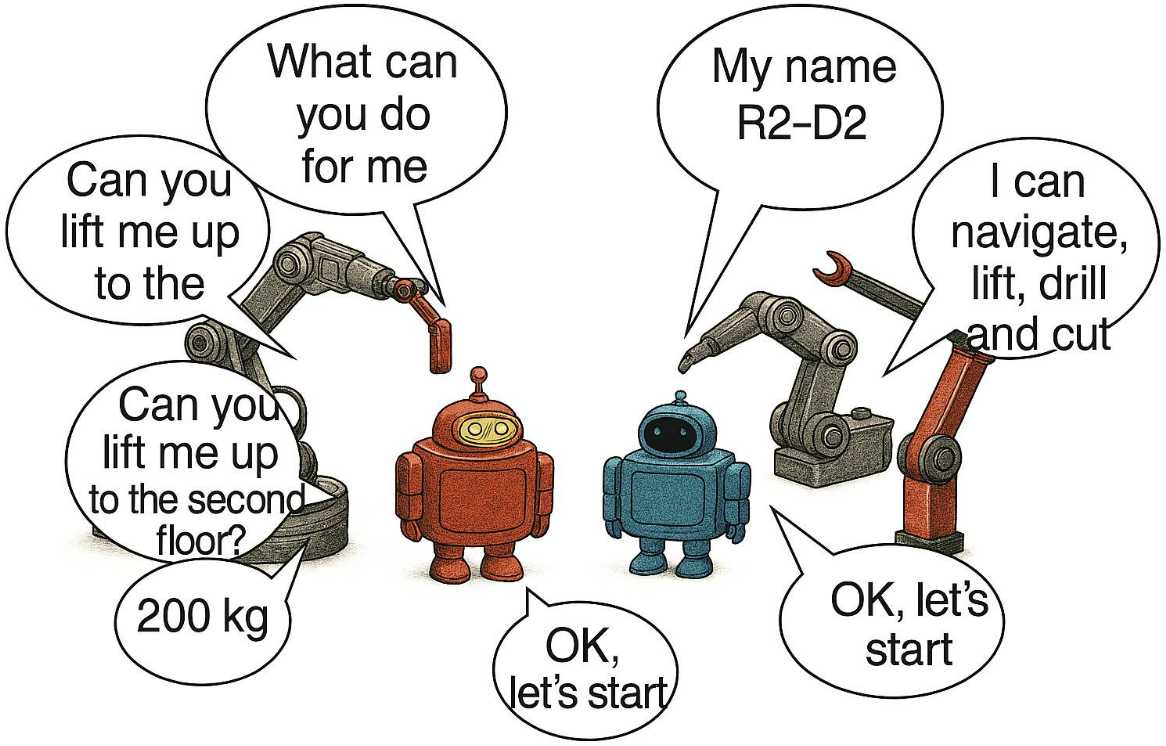
Quick OPC UA and IJT Overview

Interoperability Use Case

Without Interoperability



With Interoperability



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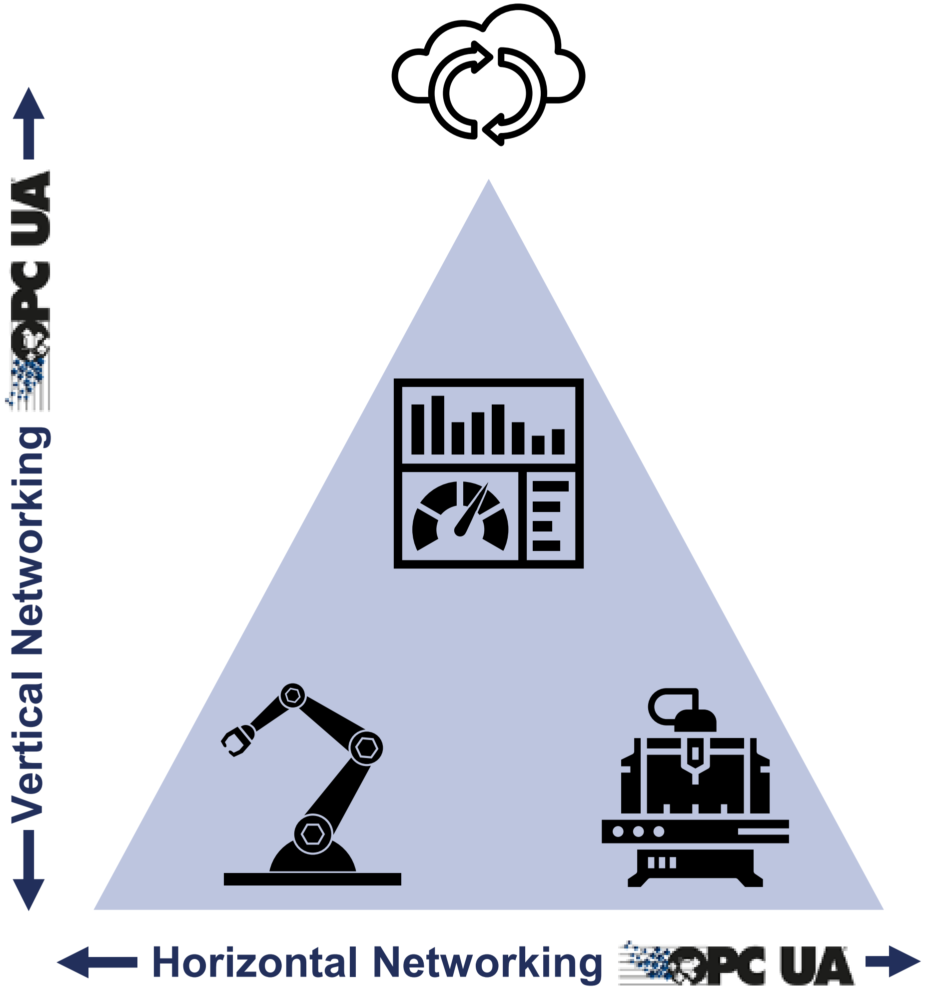

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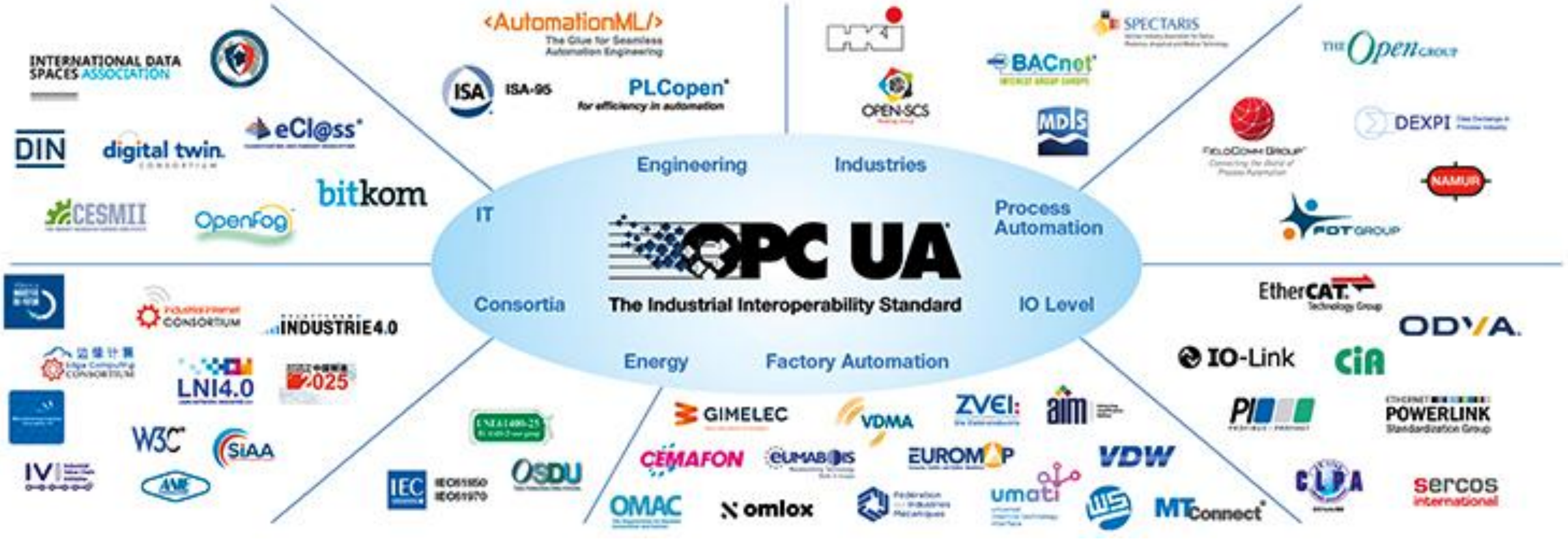
PLUS – Discovery and Security

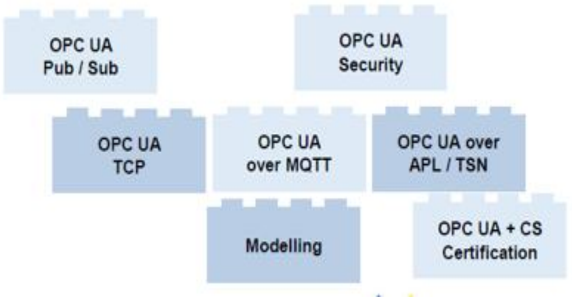
Collaboration Domain Specific Information Models

The OPC Foundation closely cooperates with organizations and associations from various branches.
Specific information models of other standardization organizations are mapped onto OPC UA and thus become portable.



Source: OPC Foundation

Interoperability Solution Summary



(OPC UA + Companion Specs) = Promise for Interoperability

- **OPC UA: Collection of technology bricks**
  - Connectivity, different protocols
  - Security
  - …

HOW to communicate
"speak the same language"

- **Companion Specifications: Collection of bricks for different markets**
  - Information modelling to describe the specific market
  - …

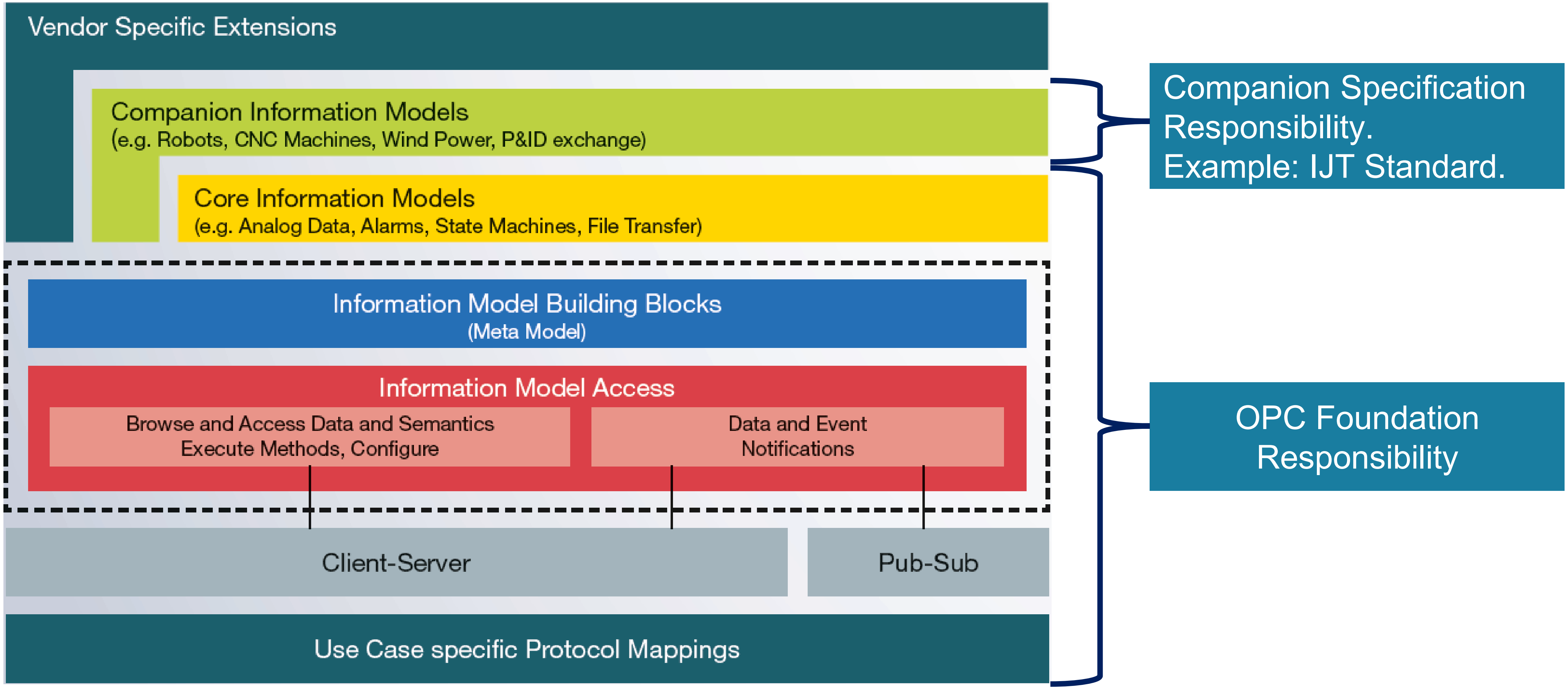
WHAT to communicate
"use the same dictionary"

- **OPC UA + Companion Specification drives towards Interoperability**
  - Mandatory bricks guarantee interoperability
  - Optional bricks allow flexibility
  - …

semantic interoperability
"understand each other"

Source: OPC Foundation

OPC UA Layered Architecture



Source: OPC Foundation

Industrial Joining Technologies (IJT) Overview

- **History**
  - May 2019 – First workshop.
- **Taxonomy**
  - https://opcfoundation.org/markets-collaboration/IJT/
- **Long term vision**
  - Started with Tightening but was interesting to see the base elements useful for other joining systems.
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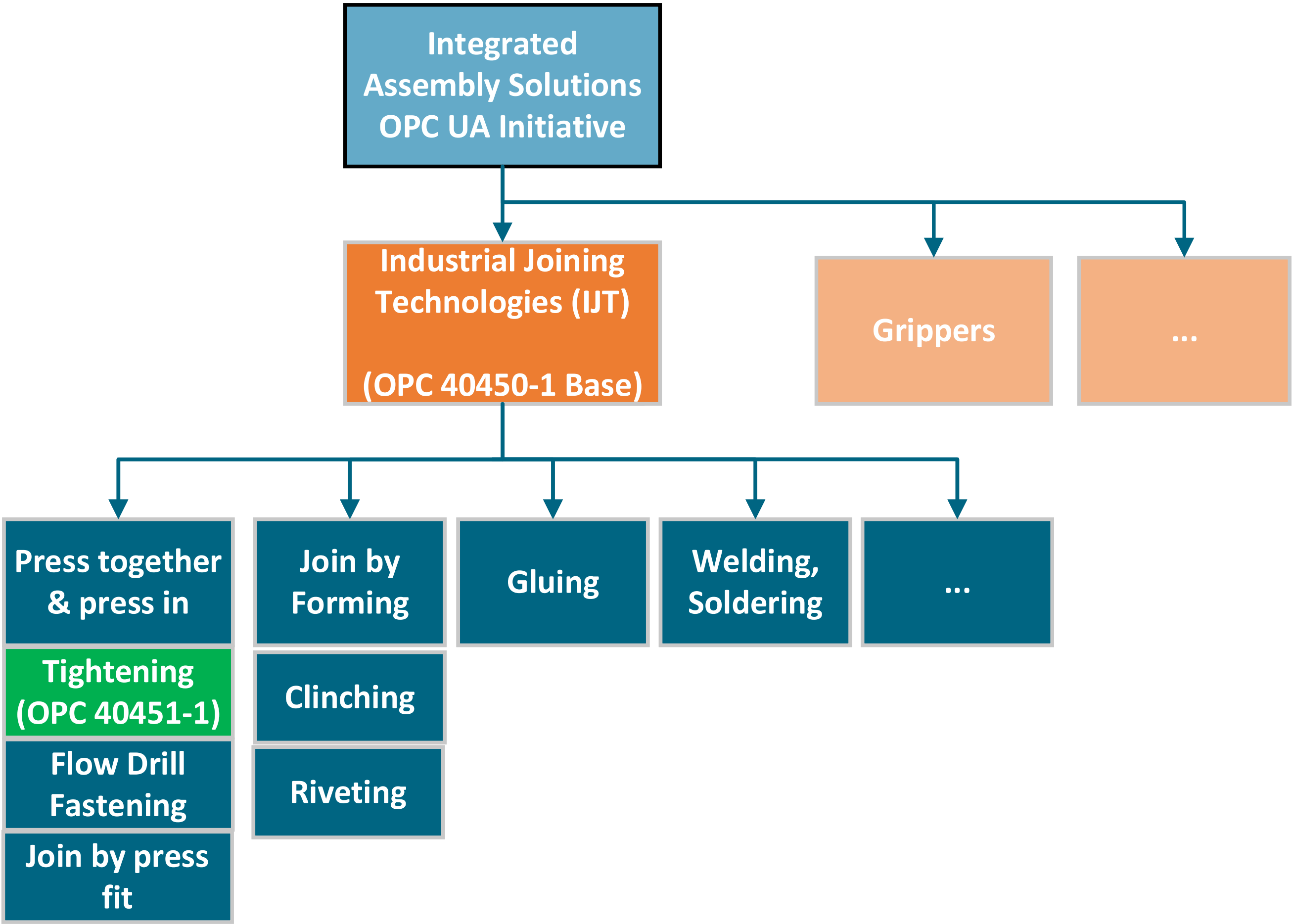

IJT Working Group Members

Atlas Copco

rexroth
A Bosch Company

WEBER

OPC
FOUNDATION

VDMA

Cleco
Production Tools

Desoutter

DEPRAG
machines unlimited

STÖGER
AUTOMATION

CSP
EFFICIENCY COMES TRUE

SCS
Concept Group

PLARAD
Torque & Tension Systems

EST
TECHNOLOGIE

AMT
Product of EST

XITASO

VISION DEVICES

MAFU
AUTOMATION

Reuse of Harmonized Specifications

NamespaceUri	Description	Use	Namespace Index	Example
http://opcfoundation.org/UA/	OPC UA Base	Mandatory	0	0:EngineeringUnits
http://opcfoundation.org/UA/DI/	OPC UA for Devices (OPC 10000-100).	Mandatory	2	2:DeviceRevision
http://opcfoundation.org/UA/AMB/	OPC UA for Asset Management Basics (OPC 10000-110).	Mandatory	3	3:IRootCauseIndicationType
http://opcfoundation.org/UA/Machinery/	OPC UA for Machinery Basic Building Blocks (OPC 40001-1).	Mandatory	4	4:MachineIdentificationType
http://opcfoundation.org/UA/Machinery/Result/	OPC UA for Machinery Result Transfer (OPC 40001-101).	Mandatory	5	5:ResultManagementType

Use Cases Overview

Asset Management

Overview and Identification of physical assets in the given system.

Example:

Manufacturer, Serial number, Software Revision, etc.

Condition Monitoring

Acquisition and processing of information that indicate the state of an asset over time.

Example:

Device Health, Temperature, etc.

Result Management

Primary process output of the joining operation.

Example:

Single Result, Batch Result, Job Result, Multi-spindle Result, etc.

Event Management

Various types of events with standard payload and filter criteria.

Example:

Tool Connected, Maintenance Events, etc.

Commands

Asset Management control mechanisms.

Joining Process Management.

Example:

Select Program, Send Identifiers, Enable Tool, etc.

Joint Management

Provides joint data associated to a Program.

Example:

Select Joint, Get Joint, etc.

Interoperability Example for OPC UA-enabled Assets

1. **Interface:** OPC UA Client/Server

2. **Data Format:** OPC UA Binary or JSON

3. **Data Model:** OPC UA Information Model

4. **Semantics:** OPC UA IJT Companion Specification

Only when all 4 things are present, we can truly understand each other!

Quick Startup with OPC UA IJT

- **Step 1:** Explore (**Example:** Around 30 minutes)
  - Download UaExpert + Run OPC UA IJT Reference Server Simulator + Browse and Get Live Data.
  - https://github.com/umati/UA-for-Industrial-Joining-Technologies
- **Step 2:** Choose type of data or Profiles (**Example:** Around 1-2 hours)
  - "Basic Joining System Server" — minimum: result access
  - "General Joining System Server" — full: results + identifiers + events
  - https://reference.opcfoundation.org/IJT/Base/v100/docs/11
- **Step 3:** Check the Client Gateway (**Example:** Around 1 day)
  - Verify your SCADA/Gateway supports OPC UA 1.05.02 Structures.
- **Step 4:** Build a Proof of Concept (**Example:** Around 2-3 days)
  - Use the IJT Console Client (Python) or IJT Web Client or IJT C# Client on the GitHub as an integration template.
- **Step 5:** Business Logic integration, Validation and Testing and Certification
  - Mapping the received data for integration and validation.
  - Plan long-term certification.

Summary

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  - **Note:** Any Release Candidate (DRAFT) is **only** accessible for an OPC Foundation Member Company. The alternative is to download from the VDMA portal: https://vdma.org/der-verband. Search the portal with the specification number. **Examples:** 40450, 40451, etc.
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OPC UA for Industrial Joining Technologies (IJT)

Topics

- Industrial Joining Technologies (IJT) Overview
- IJT Data Models
- OPC UA Profiles, Test Cases and Certification
- Demonstration and IJT Reference Applications
- OPC UA Providers
- Future Models and Extensions
- Open Discussion
- Summary
- Conclusion

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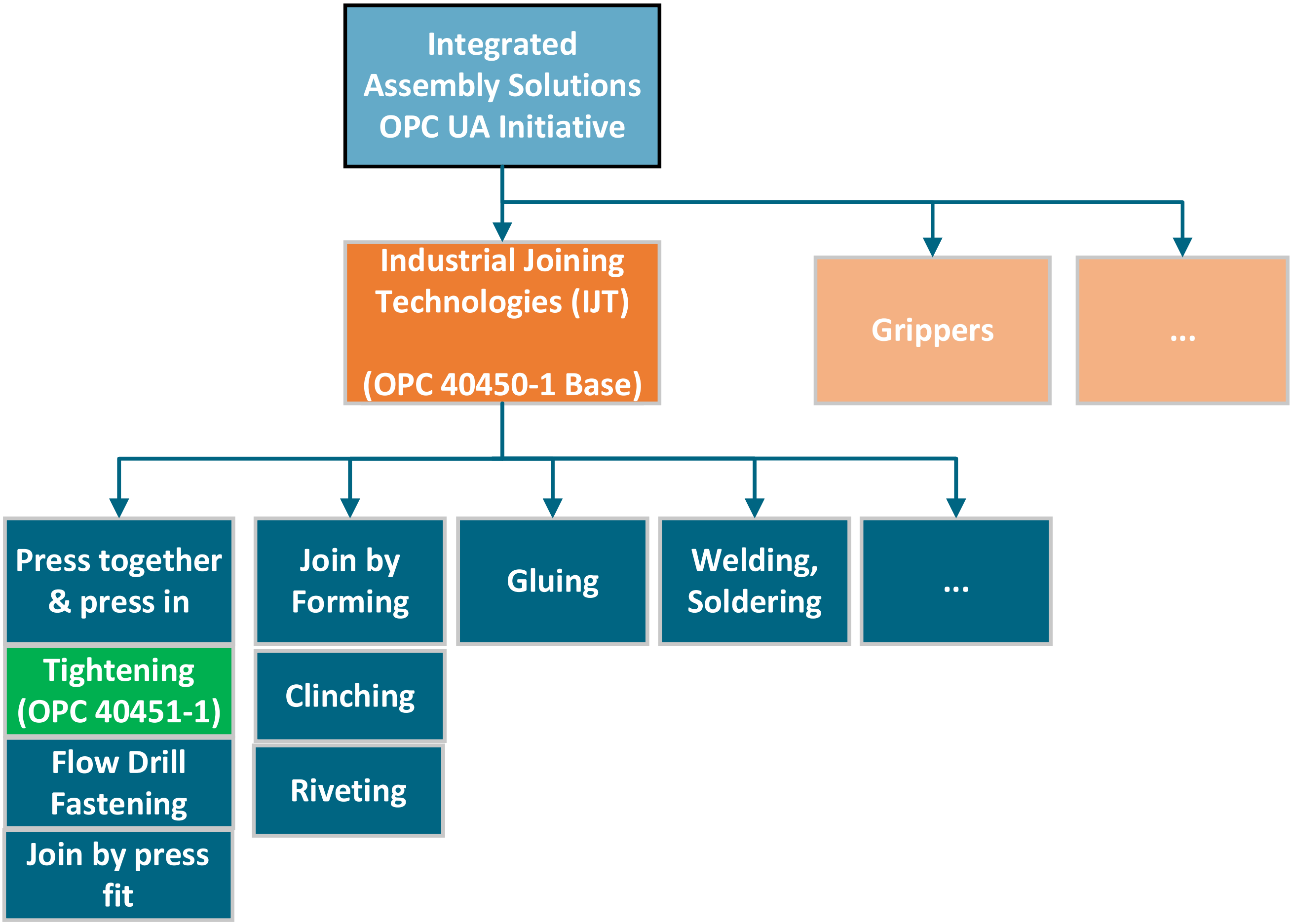

IJT Working Group Members

Atlas Copco

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EST
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VISION DEVICES

MAFU
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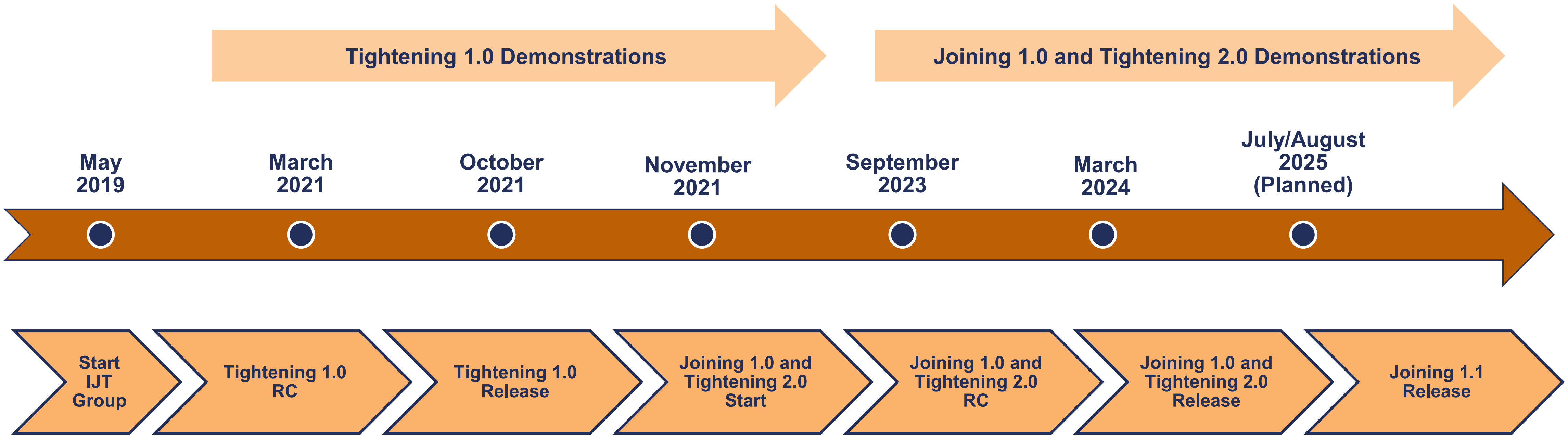
History and Milestones



Reuse of Harmonized Specifications

NamespaceUri	Description	Use	Namespace Index	Example
http://opcfoundation.org/UA/	OPC UA Base	Mandatory	0	0:EngineeringUnits
http://opcfoundation.org/UA/DI/	OPC UA for Devices (OPC 10000-100).	Mandatory	2	2:DeviceRevision
http://opcfoundation.org/UA/AMB/	OPC UA for Asset Management Basics (OPC 10000-110).	Mandatory	3	3:IRootCauseIndicationType
http://opcfoundation.org/UA/Machinery/	OPC UA for Machinery Basic Building Blocks (OPC 40001-1).	Mandatory	4	4:MachineIdentificationType
http://opcfoundation.org/UA/Machinery/Result/	OPC UA for Machinery Result Transfer (OPC 40001-101).	Mandatory	5	5:ResultManagementType

Use Cases Overview

Asset Management

Overview and Identification of physical assets in the given system.

Example:

Manufacturer, Serial number, Software Revision, etc.

Condition Monitoring

Acquisition and processing of information that indicate the state of an asset over time.

Example:

Device Health, Temperature, etc.

Result Management

Primary process output of the joining operation.

Example:

Single Result, Batch Result, Job Result, Multi-spindle Result, etc.

Event Management

Various types of events with standard payload and filter criteria.

Example:

Tool Connected, Maintenance Events, etc.

Commands

Asset Management control mechanisms.

Joining Process Management.

Example:

Select Program, Send Identifiers, Enable Tool, etc.

Joint Management

Provides joint data associated to a Program.

Example:

Select Joint, Get Joint, etc.

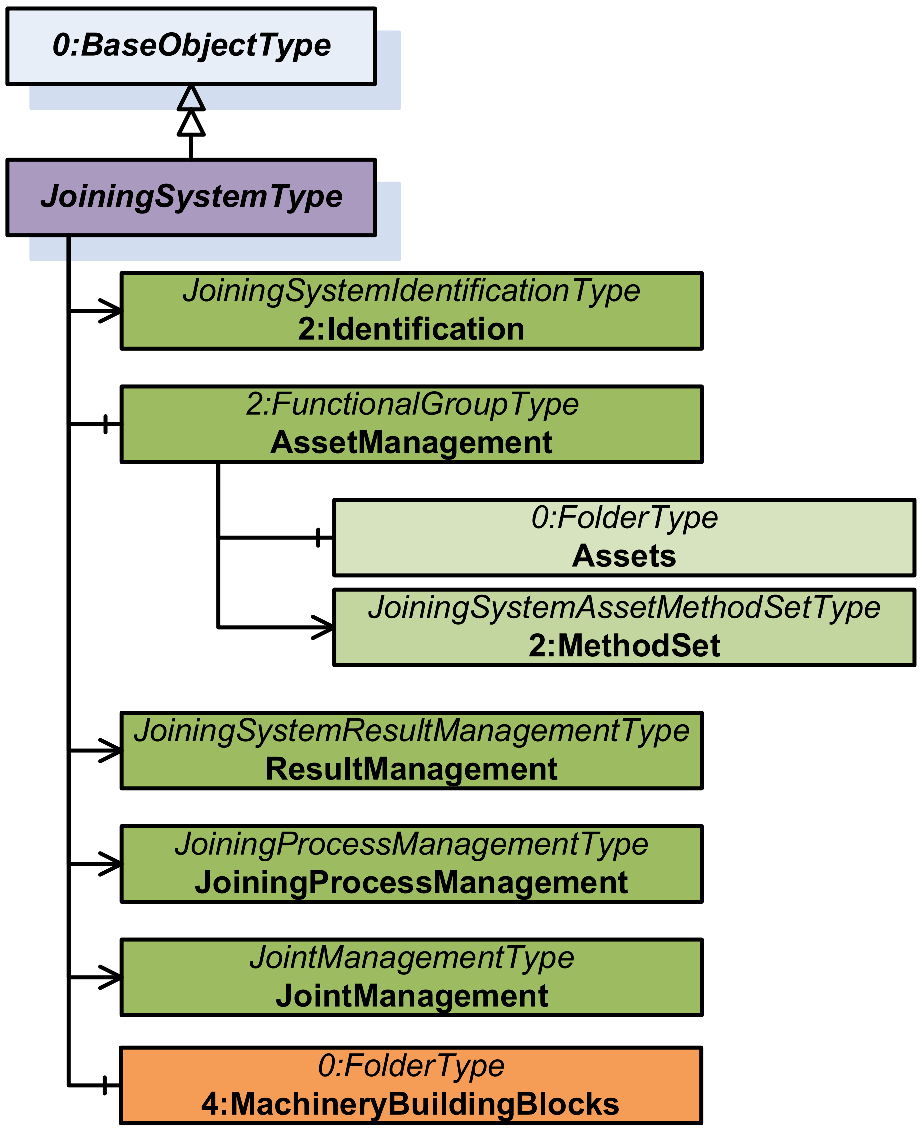
Business Case - Why OPC UA IJT?

- **WITHOUT OPC UA and IJT CS**
  - Several device vendors = several **custom** adapters to build and maintain.
  - No **consistent** and standard encryption, access control = CRA **non-compliance** risk from 2027.
  - Change device protocol = **rewrite** all integrations.
- **WITH OPC UA and IJT CS**
  - 1 standard interface → any certified device connects to any certified system.
  - TLS, certificate management, user/role management **built in** → CRA compliant by design.
  - Change device = no **additional** integration **rework**.
- **Long Term Benefits**
  - **Reduction** in **integration** cost over 5 years.

Data Models

Joining System Overview

- Standard entry point for a joining system.
- Common interface for any joining system.
- Top-level structure with building blocks for the **use cases**:
  - Asset Management
  - Result Management
  - Joining Process Management
  - Joint Management
  - …



Asset Management

Asset Management Overview

- Definition of assets building blocks.

- Diverse systems – Flexible asset management model to build different systems.
  - Fixtured, Handheld, Pneumatic, Multi-Channel, Single Channel

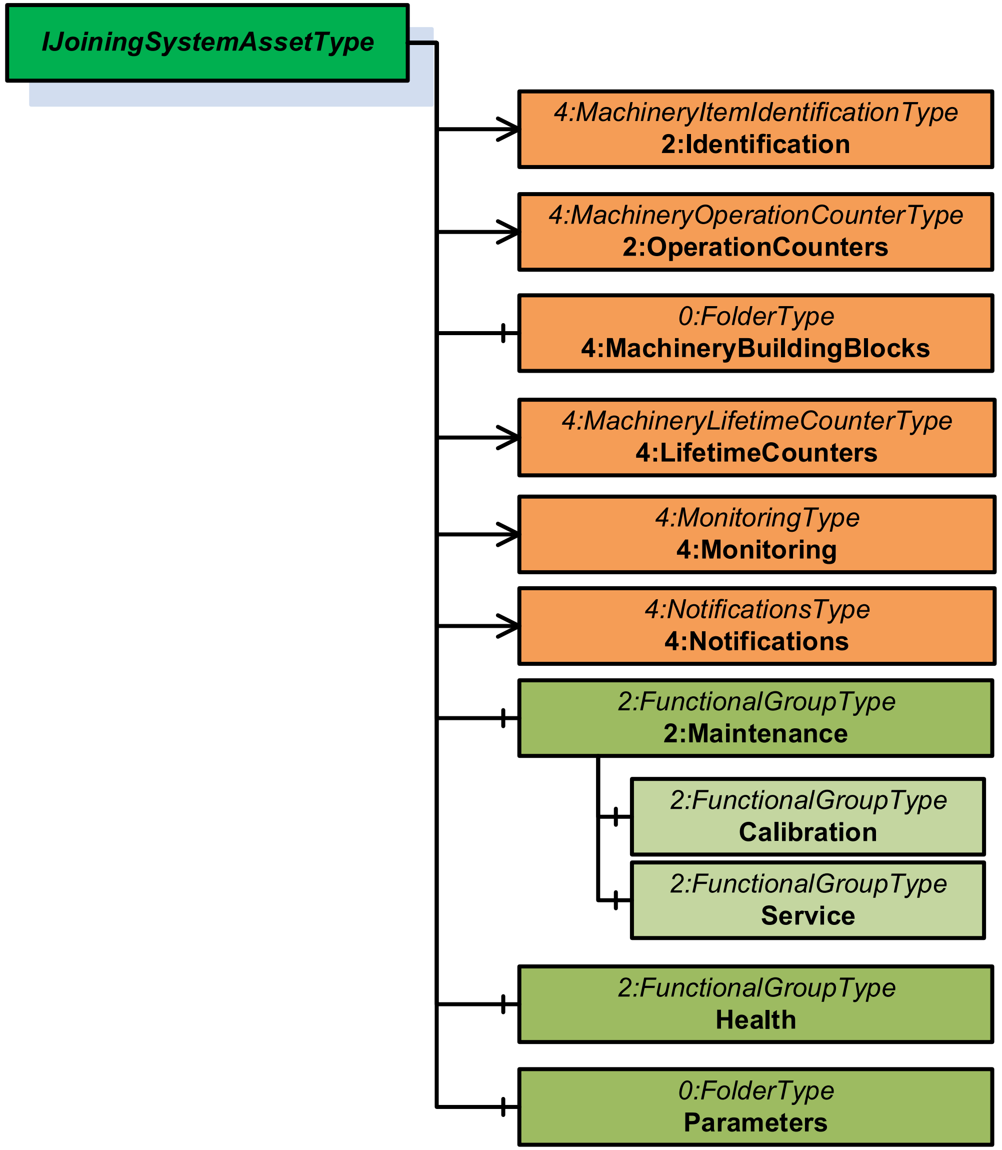
- Reuse of Machinery Building Blocks and Asset Management Basics.

- Future Extensibility with the usage of Interfaces and Add-Ins instead of concrete types.

Joining System

Controller	Tool	Servo	Feeder	Battery	Software
Accessory	Memory Device	Cable	Power Supply	Sub-Component	Virtual Station

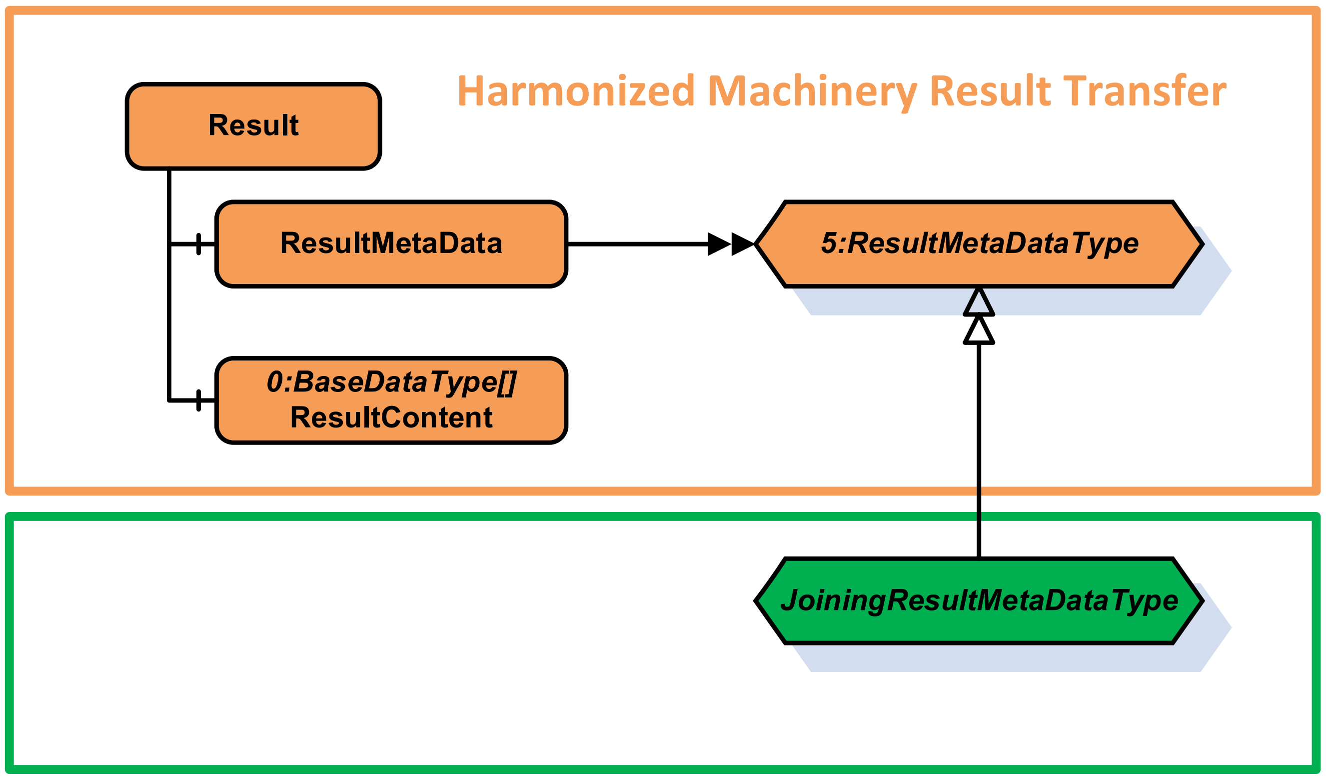
Common Asset Data



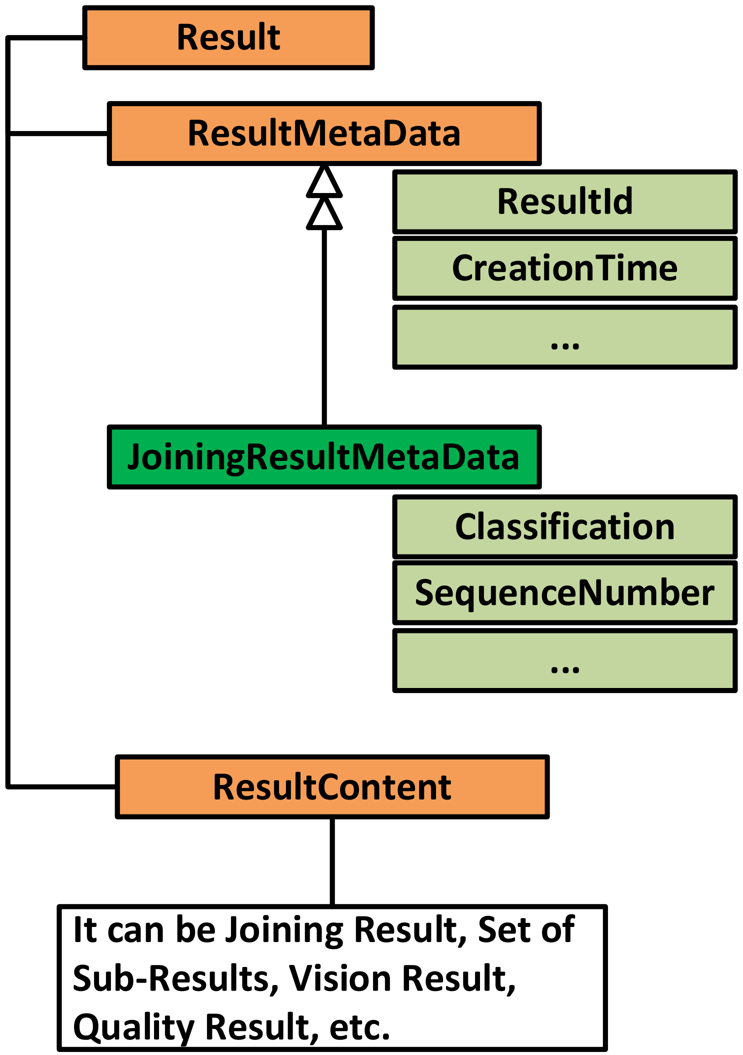
Result Management

Result Overview

Structure

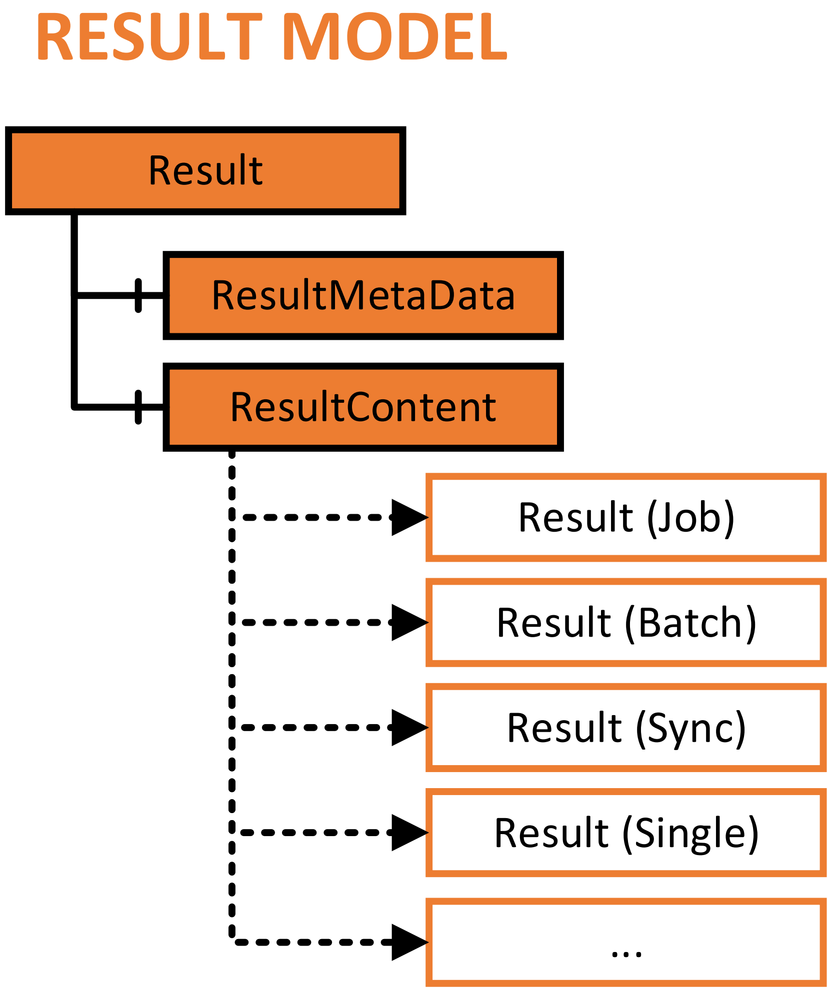


Example



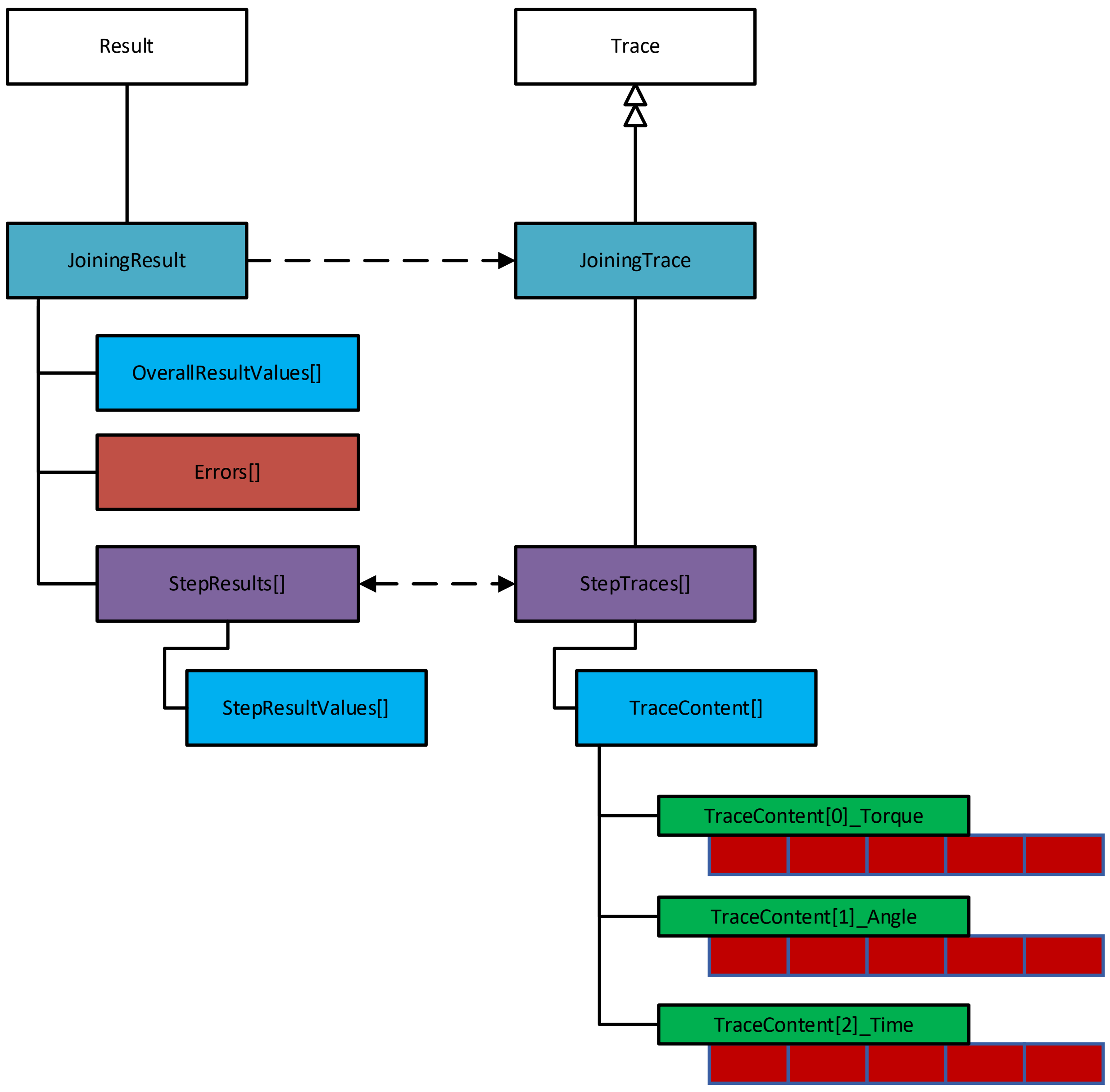
Result Classification – Multiple Use Cases

- Result
  - Common Joining Meta Data
- Result Content
  - Single Result
  - Batch Result
  - Job Result
  - Sync (Multi-spindle) Result
  - Stitching Result

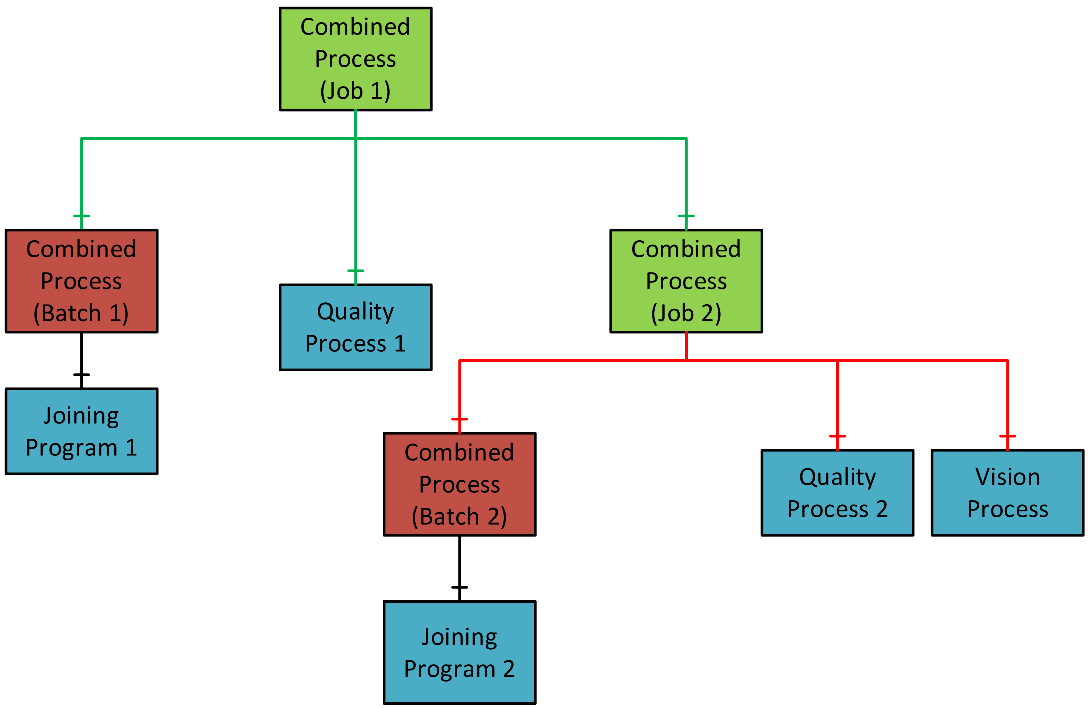


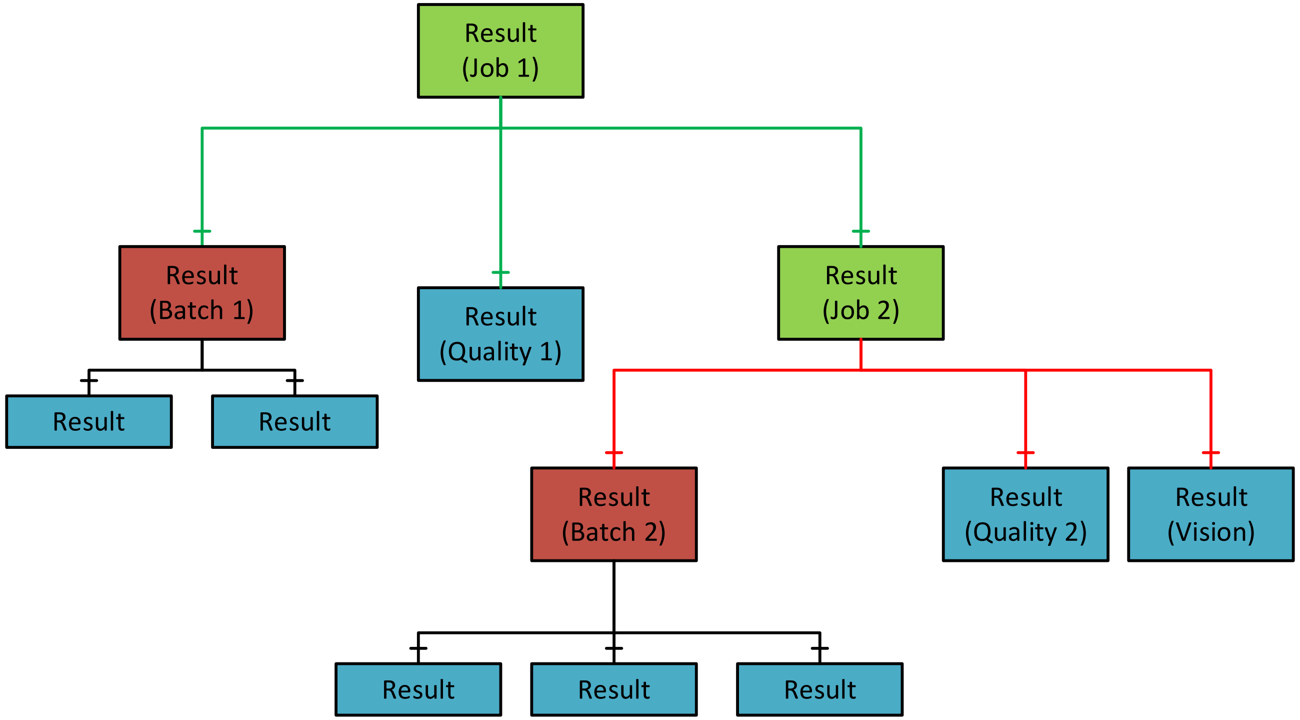
Single Joining Result Content

- Joining Result
    - Global values
    - Step Results
    - Errors
    - Traces



Combined Results – Job Result Example



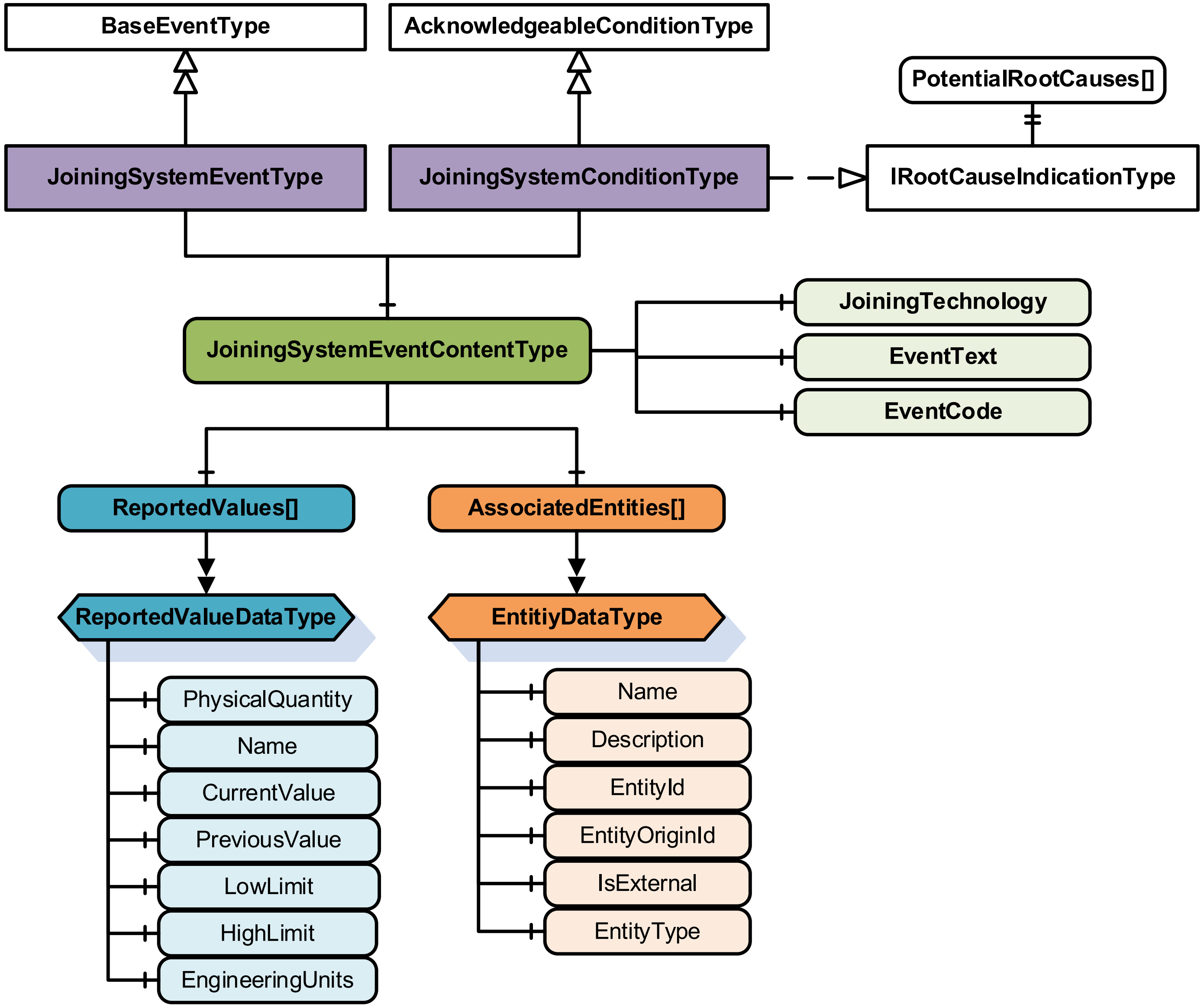


Event Management

Event Management

- **Event**
  - Simple transient information sent from the underlying system.
  - Fire and forget from the sender's perspective.

- **Condition**
  - Have a State associated.
  - Can be acknowledged by the client.



Condition Classes

- **Condition Classes**
  - Reuse from base specifications.
- **Condition Sub Classes**
  - Defined in the IJT Working Group.

- **Example 1: Tool Disconnected**
  - **ConditionClass** = SystemConditionClass
  - **ConditionSubClass** = AssetDisconnectedConditionClass

- **Example 2: The tool is out of range from the station.**
  - **ConditionClass** = SystemConditionClass
  - **ConditionSubClass** = LocationOutOfZoneConditionClass

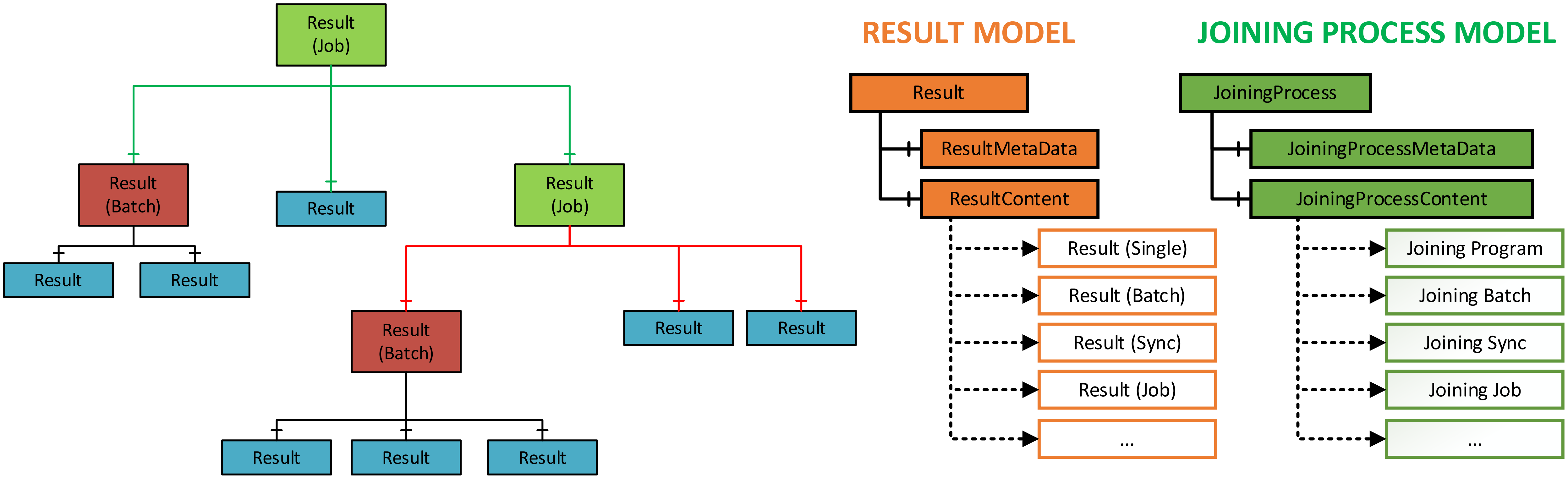
- **Example 3: Software Expired**
  - **ConditionClass** = SystemConditionClass
  - **ConditionSubClass[0]** = SoftwareConditionClass
  - **ConditionSubClass[1]** = ExpiredConditionClass

Condition Classes in Base Specifications	Condition Classes in IJT Specification(s)
OPC UA Base	AssetConnectedConditionClass
BaseConditionClass	AssetDisconnectedConditionClass
ProcessConditionClass	AssetEnabledConditionClass
MaintenanceConditionClass	AssetDisabledConditionClass
SystemConditionClass	ThresholdViolationConditionClass
SafetyConditionClass	ThresholdViolationResolvedConditionClass
HighlyManagedAlarmConditionClass	JoiningSystemUserLoggedInConditionClass
TrainingConditionClass	JoiningSystemUserLoggedOutConditionClass
StatisticalConditionClass	LocationInRangeConditionClass
TestingConditionClass	LocationOutOfRangeConditionClass
Asset Management Basics	AssetLocationConditionClass
ConnectionFailureConditionClass	DataValidationFailureConditionClass
OverTemperatureConditionClass	InputValidationFailureConditionClass
CalibrationDueConditionClass	ConfigurationChangeConditionClass
SelfTestFailureConditionClass	ErrorConditionClass
FlashUpdateInProgressConditionClass	SoftwareConditionClass
FlashUpdatedFailedConditionClass	HardwareConditionClass
BadConfigurationConditionClass	CertificateConditionClass
OutOfResourcesConditionClass	LicenseConditionClass
OutOfMemoryConditionClass	MissingEntityConditionClass
InspectionConditionClass	ExpiredEntityConditionClass
ExternalCheckConditionClass	InvalidEntityConditionClass
ServicingConditionClass	IncompatibleEntityConditionClass
ImprovementConditionClass	AcceptedEntityConditionClass
RepairConditionClass	RejectedEntityConditionClass
	AddedEntityConditionClass
	UpdatedEntityConditionClass
	RemovedEntityConditionClass
	ReceivedEntityConditionClass
	SelectedProcessConditionClassType
	SelectedEntityConditionClass
	UnacknowledgedResultsConditionClass
	EntityExpiryWarningConditionClass
	StartedEntityConditionClass
	StoppedEntityConditionClass
	NotAvailableEntityConditionClass
	NotSupportedEntityConditionClass

Joining Process Management

Joining Process

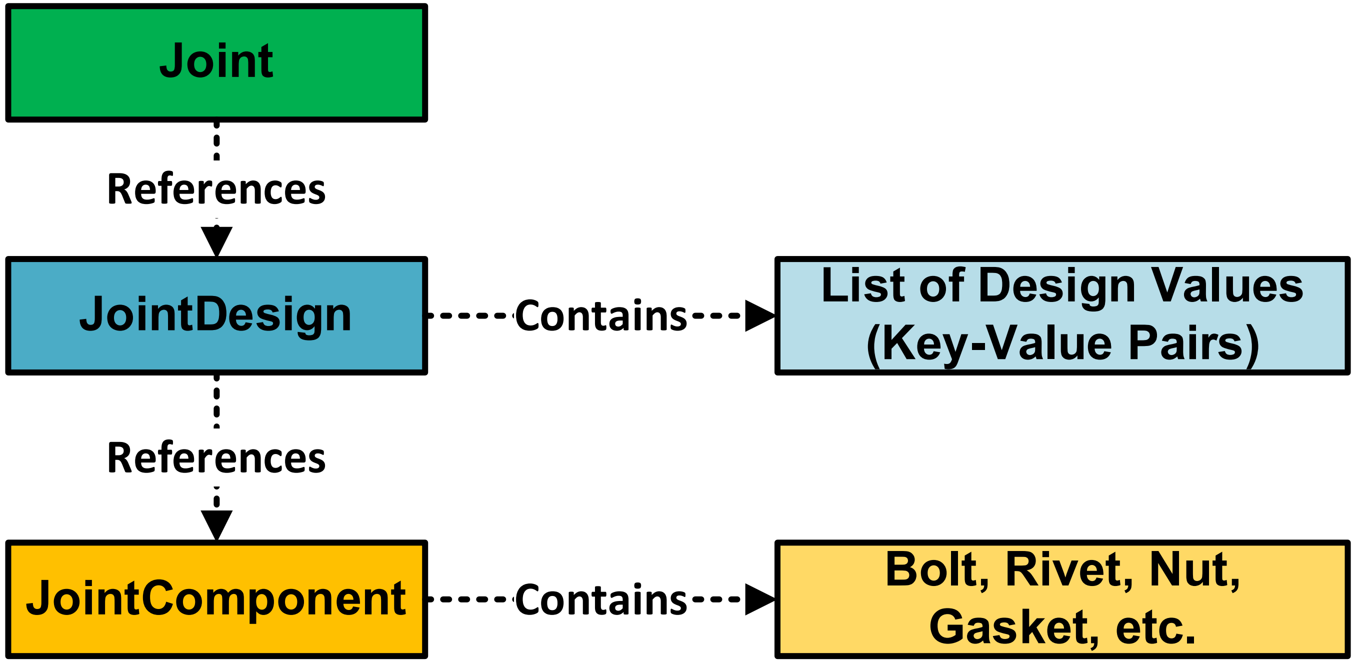
- Generic container for any type of Joining Process.
  - **Examples:** Joining Program, Joining Batch, Joining Job, etc.
- A concrete definition of the process is vendor-specific.
- The specification defines the interface to access the required processes.



Joint Management

Joint Management

- **Use Case:** The user could send information on **what** type of joint needs to be done instead of **how** the joining operation should be executed.
- Definition of Bolt, Rivet is vendor-specific.



Joining System Method List (Commands)

Joining System Method List

Asset Management	Result Management	Joining Process Management	Joint Management
SetCalibration	GetLatestResult	SendJoiningProcess	SendJoint
EnableAsset	GetResultById	GetJoiningProcess	SendJointDesign
DisconnectAsset	GetResultIdListFiltered	GetJoiningProcessList	SendJointComponent
RebootAsset	ReleaseResultHandle	GetJoiningProcessRevisionList	GetJointList
SendFeedback	AcknowledgeResults	SetJoiningProcessMapping	GetJointRevisionList
GetFeedbackFileList	RequestResults	SelectJoiningProcess	GetJointDesignList
SetOfflineTimer	RequestUnacknowledgedResults	DeselectJoiningProcess	GetJointComponentList
SetTime		IncrementJoiningProcessCounter	GetJoint
SendIOSignals		DecrementJoiningProcessCounter	GetJointDesign
GetIOSignals		SetJoiningProcessCounter	GetJointComponent
SendIdentifiers		SetJoiningProcessSize	SelectJoint
SendTextIdentifiers		ResetJoiningProcess	DeleteJoint
GetIdentifiers		AbortJoiningProcess	DeleteJointDesign
ResetIdentifiers		StartJoiningProcess	DeleteJointComponent
ExecuteOperation		StartSelectedJoining	
GetErrorInformation		DeleteJoiningProcess	
		GetSelectedJoiningProgram	

Status

- OPC **40450-1** UA CS for **Joining** Systems
  - **Link:** https://reference.opcfoundation.org/nodesets?u=http://opcfoundation.org/UA/IJT/Base/
  - **1.00.0:** Published in March 2024.
  - **1.01.0:** Published in October 2025.
- OPC **40451-1** UA CS for **Tightening** Systems **2.00.0**
  - **Link:** https://reference.opcfoundation.org/nodesets?u=http://opcfoundation.org/UA/IJT/Tightening/
  - **1.00.0:** Published in October 2021.
  - **2.00.0:** Published in March 2024.
  - **2.00.1:** Published in October 2025.
- IJT Working Group Current Progress
  - Test Cases definition.
  - Exploring of new use cases and issues/feedback in the IJT CS.

OPC UA Profiles and Certification

OPC UA Profiles and Certification

- **What is OPC Certification?**
  - https://opcfoundation.org/certification/overview-benefits/
  - https://opcfoundation.org/products/?certified=yes
- **Compliance Test Tools (CTT)**
  - OPC Foundation provides tools which can validate OPC UA Core functionality.
  - https://opcfoundation.org/developer-tools/certification-test-tools
- **Companion Specification Certification**
  - OPC Foundation provides support in testing CS manually or also helps in defining the automated test cases based on collaboration.
  - **Working Group Responsibility:** Definition of Profiles, CUs, Facets, and Test Cases.
  - **Certification Responsibility:** OPC Foundation Certification Lab.
- **OPC UA Profiles**
  - Terms and Definitions: https://reference.opcfoundation.org/Core/Part7/v105/docs/3.1
  - Profile reporting tool: https://profiles.opcfoundation.org/

IJT Profiles Overview

IJT Profiles Overview

- OPC UA Specifications define a set of rules that are needed for a product to be certified.
  - It is done using the definition of Conformance Units, Facets and Profiles.

- The following image provides an overview of profiles defined in IJT specifications.

CU/Facet	Facet/Profile
Joining System Base	Basic Joining System Facet
Result Server Facet	
Asset Management Assets Server Facet	
Basic Joining System Facet	General Joining System Facet
Joining Result Server Facet	
Trace Server Facet	
Identifiers Methods Server Facet	
Event Management Server Facet	
Joining Process Base Server Facet	
Result Content	
Result Internal Identifiers	
Result External Identifiers	
Method Input Argument	

Independent Selectable Features
RESULTS
Batch Result Server Facet
Sync Result Server Facet
Stored Result Server Facet
Job Result
Partial Consolidated Result
Self Contained Consolidated Result
Consolidated Result with references
Result Value FINAL Tag
ASSET METHODS
Asset Connection Server Facet
Enable Tool Server Facet
JOINTS
Joint Server Facet
Joint Design Server Facet
Joint Component Server Facet
JOINING PROCESS
General Process Operations Server Facet
Sequential Process Operations Server Facet
Start Joining Process
MISC.
Engineering Units

Test Cases and Compliance - Certification

- IJT CS has **defined** modular **conformance units** and **profiles**.

- IJT Group working on the **Test Cases** based on the standard approach and template from OPC Foundation.
  - After the test cases, **automated** test scripts for **certification** will be **planned** and **designed** with the help of OPC Foundation.
  - The Test Cases will be published in OPC Foundation portal once reviewed and approved.
    - https://profiles.opcfoundation.org/v104/Reporting/
    - **Planned** target to have DRAFT Test Cases: Q2/Q3 2026.

- **Certified Product** = Plug-and-Play with ANY certified application from any vendor.

Demonstrations and IJT Reference Applications

Prototypes / Reference Implementations

- **Main GitHub:**
  - https://github.com/umati/UA-for-Industrial-Joining-Technologies
- **OPC UA Generic Client (UaExpert):** LINK
  - This the **primary generic** test client which is free of charge and can be used to understand any OPC UA concepts.
- **IJT Server Simulator:** LINK
  - The functionality in the simulator is like the OPC UA Adapter on a controller. Refer to the document in the above link for usage instructions.
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- **IJT C# Client:** LINK
  - It is a simple C# Client based on OPC Foundation C#.NET SDK for example usage.
- **IJT Test Client:** LINK
  - It is a base test client for testing OPC UA IJT Server features. It is a base testing framework to start with testing of the OPC UA IJT Server features and has more features than IJT Console Client.
- **YouTube Recording:** IJT_YouTube_Recording
  - It is the global webinar done via VDMA IJT Group in **2024** on OPC UA, IJT Overview and Models along with demonstration.

Demonstrations

- Demonstrations are done using umati infrastructure.

- **What is umati?**
  - It is a common dashboard connected to several machines from various vendors using a standard information model.
  - Refer to https://umati.org/ for more information.

- **Demonstration:**
  - https://umati.app/
  - Done in automatica 2022, 2023 and 2025.

OPC UA Providers

OPC UA Providers – UA 1.05.02 Structures Support

- IJT Group followed up with several OPC UA Providers regarding support of OPC UA Structures/Extension Objects and most of the common OPC UA providers have added required support in the past 2 years.
    - Commercial
        - Matrikon
        - Unified Automation
        - Prosys OPC UA Forge
        - Ignition (Inductive Automation)
        - Kepware (PTC) – In Progress
        - …
    - Open Source
        - Python opcua-asyncio
        - NodeOPCUA – In Progress
        - …
    - How to Get Started?
        - **GitHub:** https://github.com/umati/UA-for-Industrial-Joining-Technologies

Future Models and Extensions

- The long-term vision to have a **common** base for various joining technologies is in place.
- **Part Management**
  - OPC Harmonization agreed to define minimal building block for Part Management in March 2026.
- **Location Models**
  - Hierarchical, Operational, Relative Spatial Location and Global Position Models exists.
  - Plan to review them and check if those needs to be integrated in the IJT CS or it is done at application level.
- **Quality Use Cases in Joining Domain**
  - Options to include quality use cases are in discussion.
- **Generic Accessory Signal Management**
  - It is under discussion with OPC UA FX and Harmonization Group.
- **Asset Management Extensions**
  - Network Data, etc. Integration of additional features from **new** Machinery Specifications as needed.
- **Feedback and Missing Use Cases**
  - Any additional commands?

Open Discussion

- **Challenges in OPC UA?**
- **Next Generation** Architecture in Plants?
- OPC UA **solves** the CRA use cases.
- OPC UA is an **enabler** for Interoperable Ecosystem.
  - In long term, with OPC UA as a standard interface reduces the maintenance of several legacy systems.
- **GitHub** Reference Applications provides multiple examples for Client implementation.
  - https://github.com/umati/UA-for-Industrial-Joining-Technologies
- **Feedback and Missing Use Cases**
  - Any additional commands?
- **Subscribe** to latest updates from IJT Group using the following Forms: **Click Here**

Greenfield vs Brownfield in Joining Domain

- For **brownfields**, a phased migration can be done. It **depends** on the use case and other factors.
  - **Example:** For control mechanisms, existing legacy protocols can be used and OPC UA can be deployed for data acquisitions.

- For **Greenfields**, it is recommended to use OPC UA with IJT Companion Standard.

- **OPC UA** and **other** protocols can **coexist** together.

Quick Startup with OPC UA IJT

- **Step 1:** Explore (**Example:** Around 30 minutes)
  - Download UaExpert + Run OPC UA IJT Reference Server Simulator + Browse and Get Live Data.
  - https://github.com/umati/UA-for-Industrial-Joining-Technologies
- **Step 2:** Choose type of data or Profiles (**Example:** Around 1-2 hours)
  - "Basic Joining System Server" — minimum: result access
  - "General Joining System Server" — full: results + identifiers + events
  - https://reference.opcfoundation.org/IJT/Base/v100/docs/11
- **Step 3:** Check the Client Gateway (**Example:** Around 1 day)
  - Verify your SCADA/Gateway supports OPC UA 1.05.02 Structures.
- **Step 4:** Build a Proof of Concept (**Example:** Around 2-3 days)
  - Use the IJT Console Client (Python) or IJT Web Client or IJT C# Client on the GitHub as an integration template.
- **Step 5:** Business Logic integration, Validation and Testing and Certification
  - Mapping the received data for integration and validation.
  - Plan long-term certification.

Few Common Questions and Answers…

- We use legacy protocols, is it mandatory to migrate?
  - No, OPC UA coexists with legacy. Start with data acquisition and migrate control layer later in phases.
- Can we get torque curves in real time?
  - Yes, JoiningResult consists of time-series values accessible via Events, Data Subscription or Methods.
- Can we get historical Results?
  - Yes
- Can we get data from different systems such as Multi-Spindle, Handheld, etc.?
  - Yes
- Can we interact with the system using commands?
  - Yes
- Do we have reference implementation examples?
  - Yes, in the main GitHub: https://github.com/umati/UA-for-Industrial-Joining-Technologies
- What is the cost?
  - OPC UA Specifications are free.
  - Many Open-Source SDKs are free.
  - Commercial SDKs with nominal cost.
  - Certification lab has a fee and first and primary need is for the OPC UA Server providers to have a IJT compliant OPC UA Server.

Summary

- **Overview of OPC Foundation, OPC UA and IJT**
  - https://opcfoundation.org/about/opc-technologies/opc-ua/
  - https://opcfoundation.org/markets-collaboration/IJT/
- **OPC UA IJT Group Presentation:** Refer to the **OPC UA IJT Group Presentation.pdf/pptx.**
  - https://github.com/umati/UA-for-Industrial-Joining-Technologies/tree/main/IJT_Documents
- **Specifications/Online Reference**
  - **Joining:** https://reference.opcfoundation.org/nodesets?u=http://opcfoundation.org/UA/IJT/Base/
  - **Tightening:** https://reference.opcfoundation.org/nodesets?u=http://opcfoundation.org/UA/IJT/Tightening/
  - **Note:** Any Release Candidate (DRAFT) is **only** accessible for an OPC Foundation Member Company. The alternative is to download from the VDMA portal: https://vdma.org/der-verband. Search the portal with the specification number. **Examples:** 40450, 40451, etc.
- **OPC UA IJT Prototypes/Reference Implementations**
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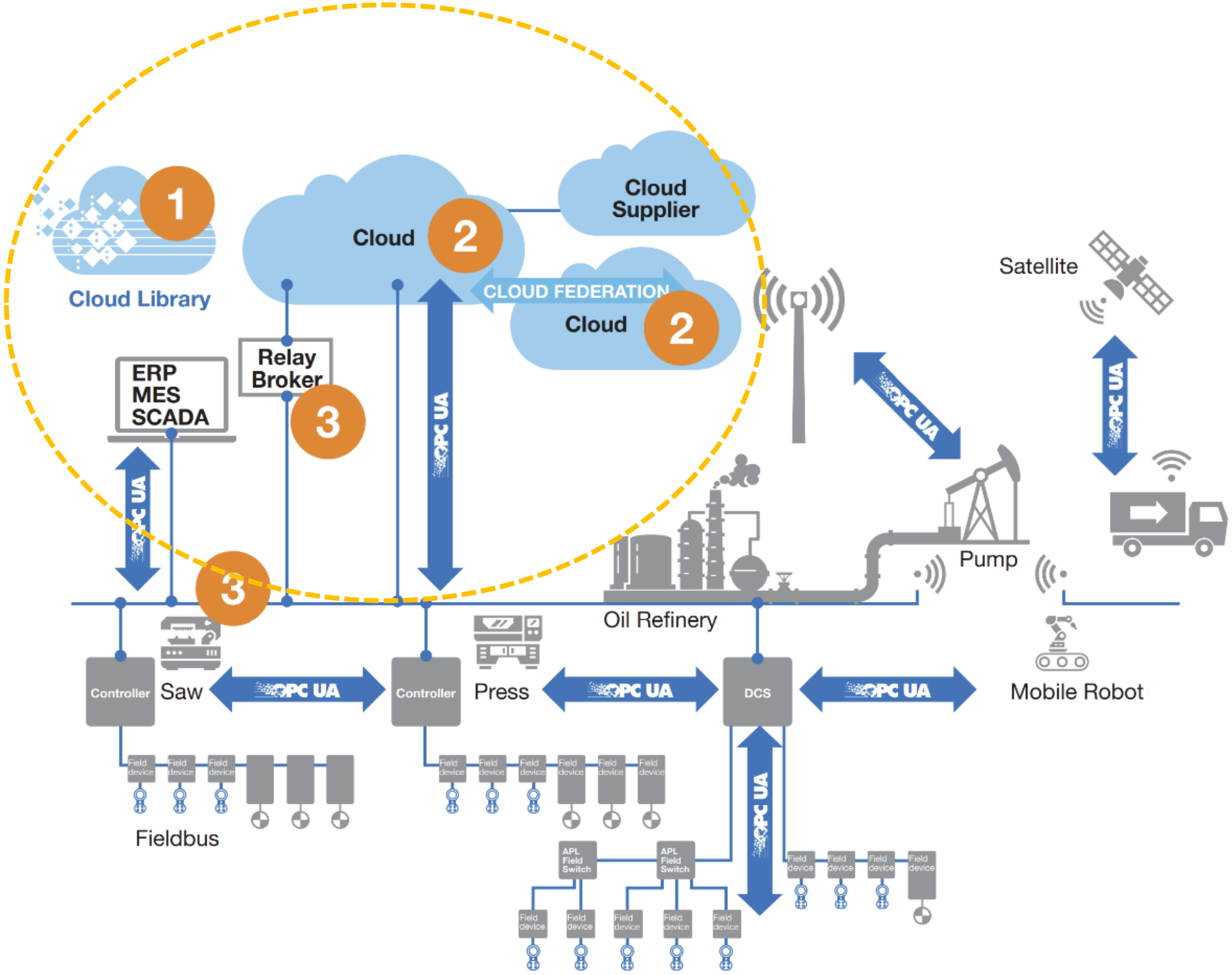
Questions?

Additional Slides

OPC Legacy and OPC UA History

- Refer to the following link which explains the Before and After OPC Scenarios:
- https://www.ia.omron.com/product/special/sysmac/nx1/opcua.html

UA Cloud Initiative

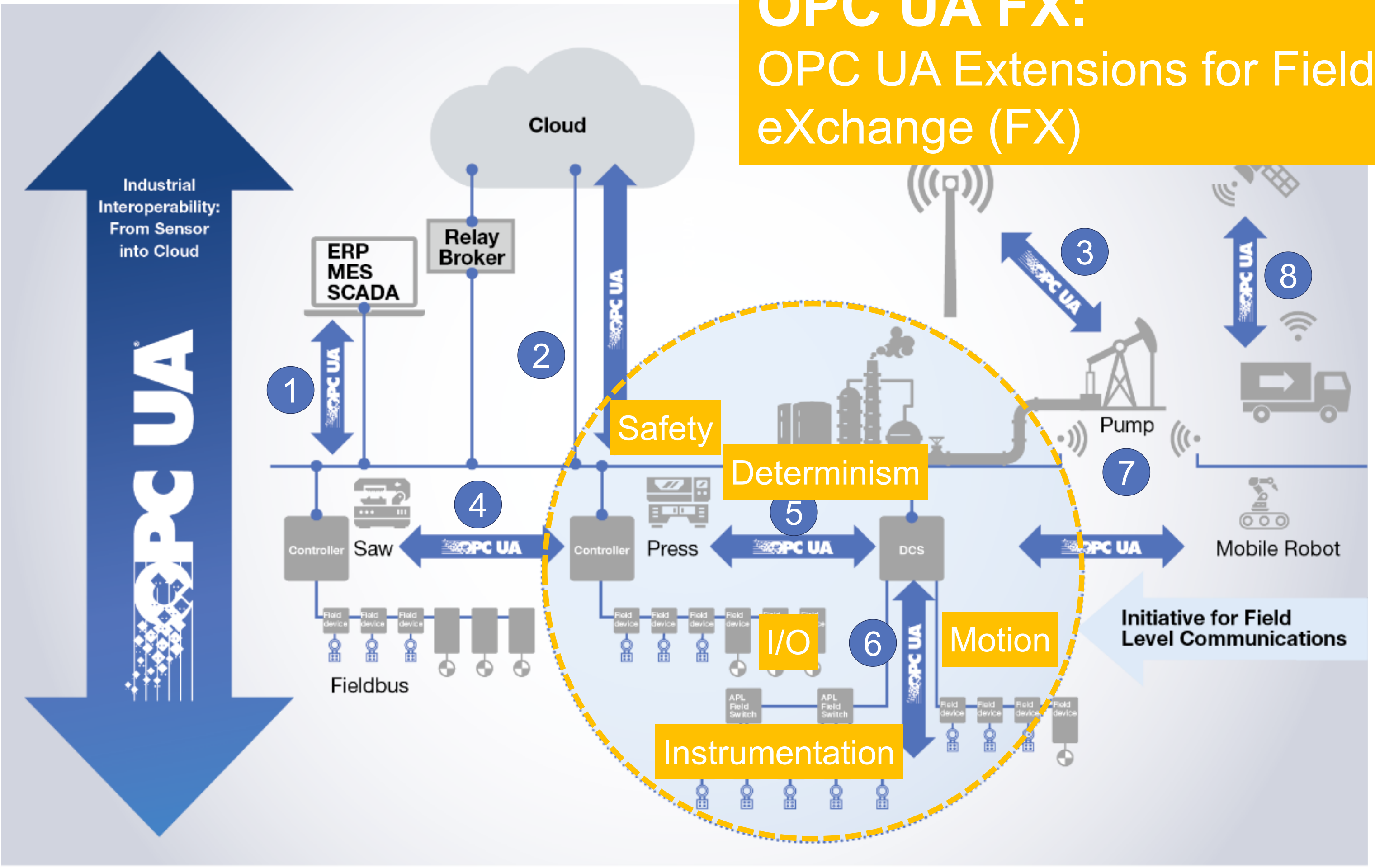


1. **Cloud Library**
   - Repository for OPC UA based information models (IMs)
   - Upload, store, search, download IMs
2. **Cloud Federation**
   - Standardized communication
   - Cloud to Cloud
3. **Asset / Edge / Cloud**
   - Standardized communication
   - Field to Cloud
   - Cloud to Field
4. **Education, IIOT Starter Kit**
   - Success stories

Source: OPC Foundation

OPC UA for Field eXchange (FX):
Extending OPC UA to the field incl. Determinism, Safety & Motion

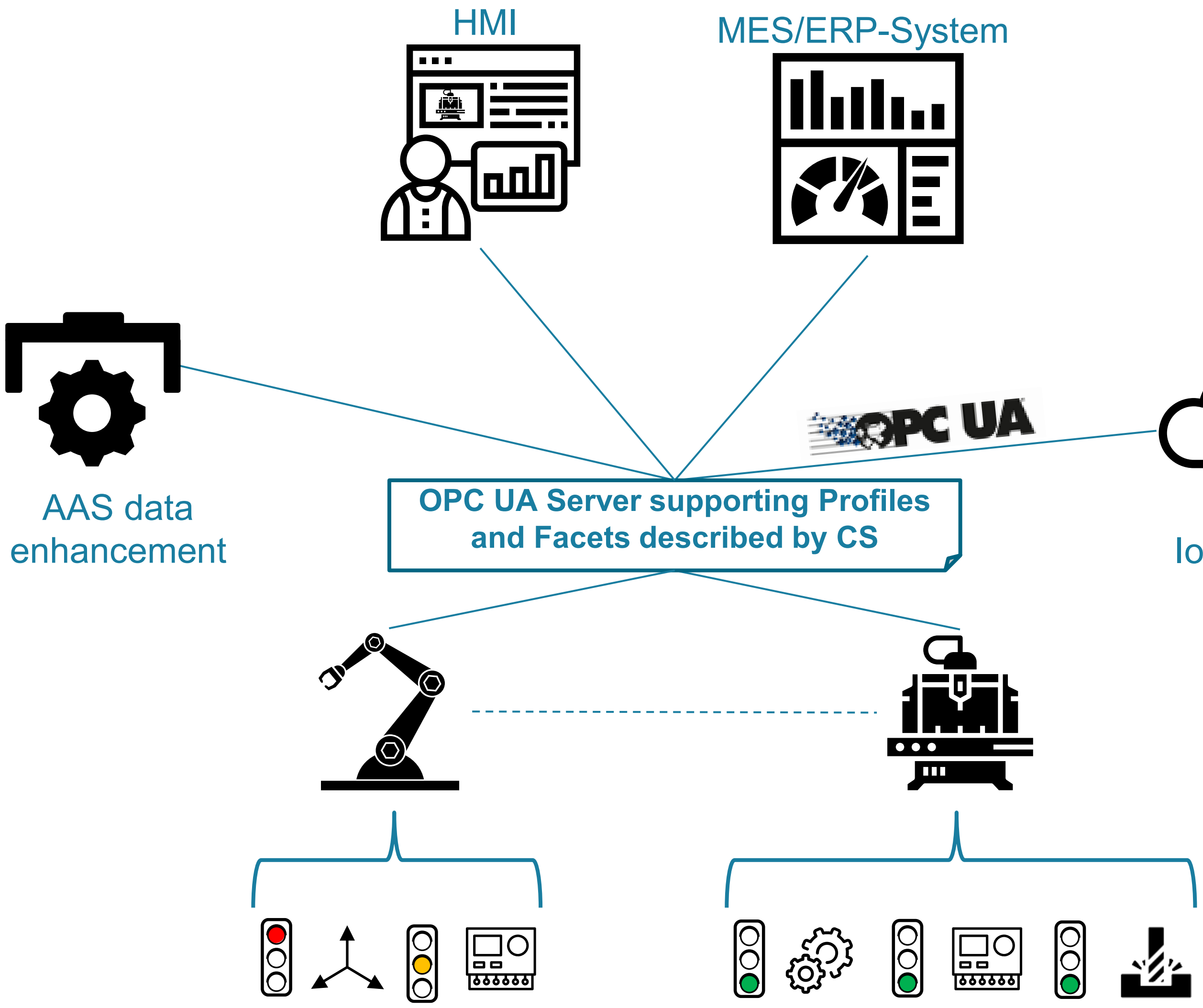
OPC UA FX:
OPC UA Extensions for Field eXchange (FX)



1. IT / OT Communication
2. Cloud Integration
3. Secure Remote Access
4. Local OT Communication
5. Controller to Controller
6. Controller to Device incl. Device to Device
7. Wireless Integration (5G)
8. Future Ready

Source: OPC Foundation

OPC UA as the standardized communication interface of machines & components



Release 1 and Release 2 Differences

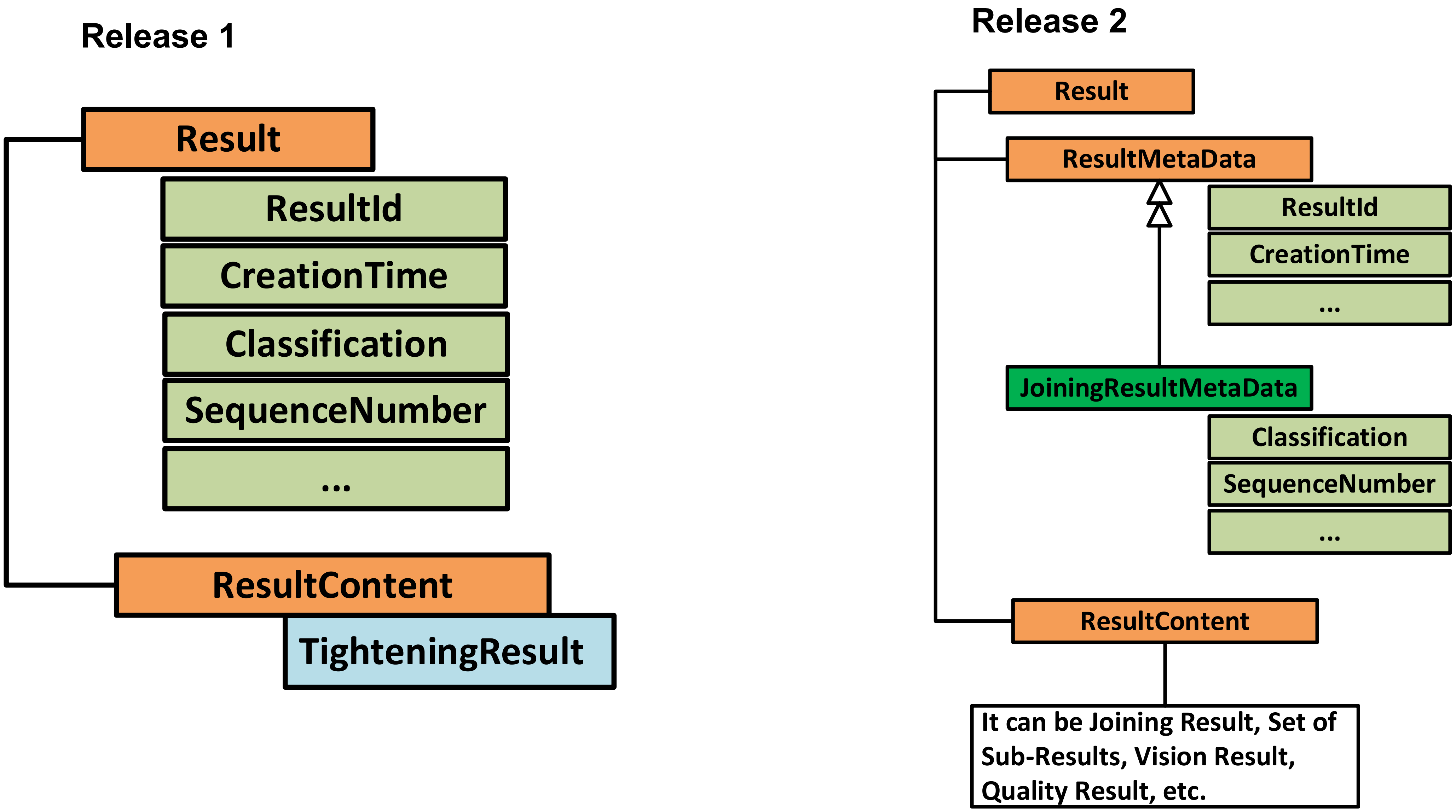
Overview

- **Release 1** was **only** for Tightening Domain.
  - OPC 40451-1 UA CS for Tightening Systems **1.00.0.**

- During **Release 2** work, most of the use cases are generalized for various joining systems. Hence, the models were **extended** and **moved** to a new **base** specification.
  - OPC 40450-1 UA CS for Joining Systems **1.00.0.**
  - Tightening Specification is upgraded to OPC 40451-1 UA CS for Tightening Systems **2.00.0.**

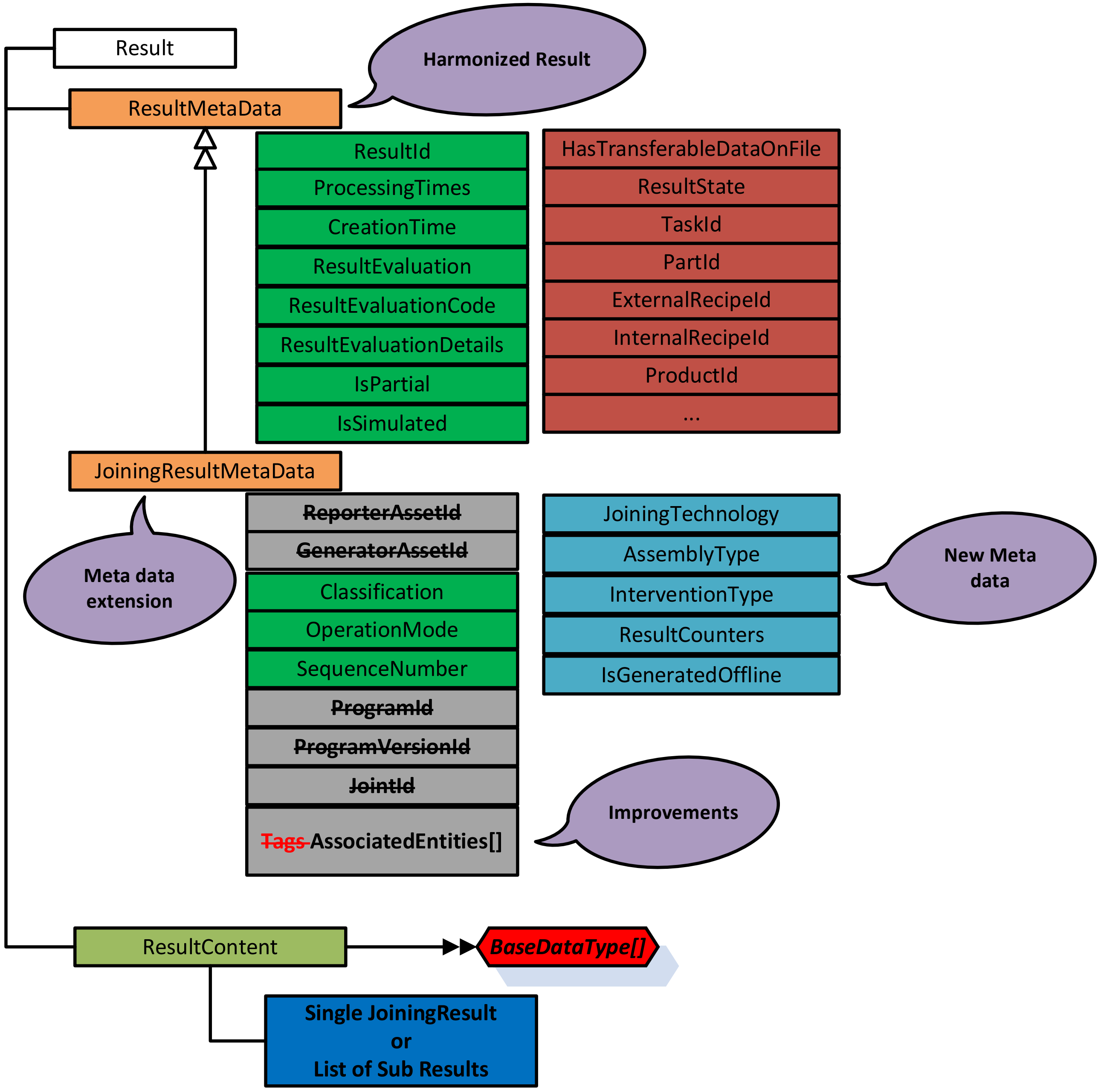
- **Why is it a major version release?**
  - Most of the models from the Tightening Specification are generalized and moved to a new base specification.

Result Structure Changes



Result Data Changes

- **Machinery Result** Usage from Harmonization Group.
  - **Result** properties **grouped** as **ResultMetaData**.
- **Additional** set of properties to cover the use cases for Batch/Job/Sync Result, etc.
- **Generalization** of **Tightening** Result to **Joining** Result.
- **Uses** UA 1.05.02 Structures.



System, Asset and Event Changes

- **Improvement** in the Joining System for Identification AddIn.
- **Extension** of Joining System to include newer models.
- **Reorganization** and **Extension** of Asset Management Model.
  - Software and Virtual Station as new assets.
- **Extension** of Event model by **common joining payload** and **filtering** mechanism.